

Applying Automotive EDR Data to Traffic Crash Reconstruction

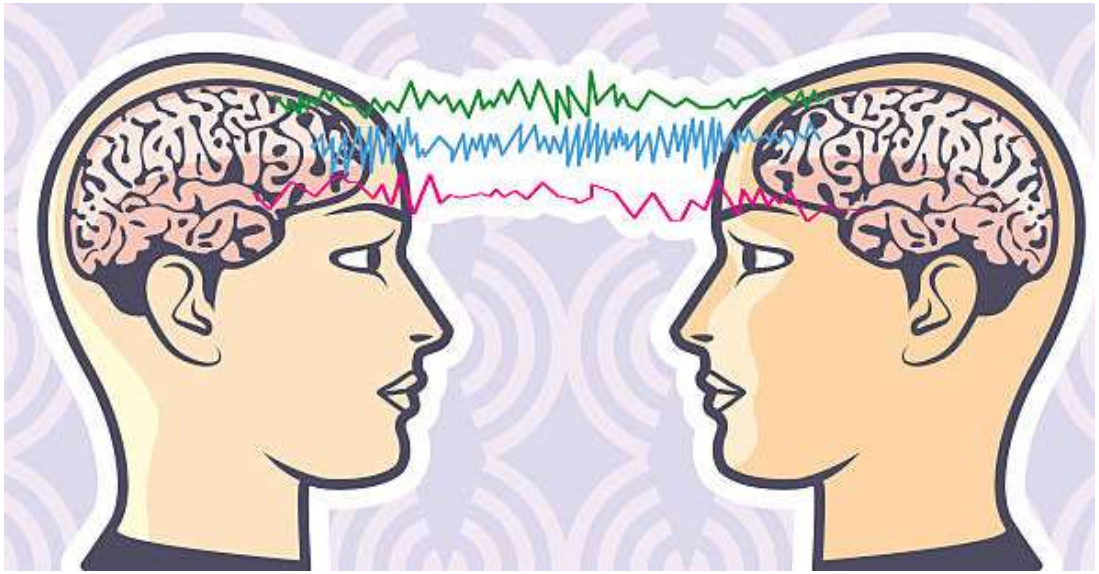
VIRTUAL

Virtual Differences from Live



In 1986, a Russian airline pilot bet with his co-pilot that he could land the plane flying blind, he curtained all the cockpit windows and crashed the plane, killing 70 out of the 94 passengers, the pilot survived.

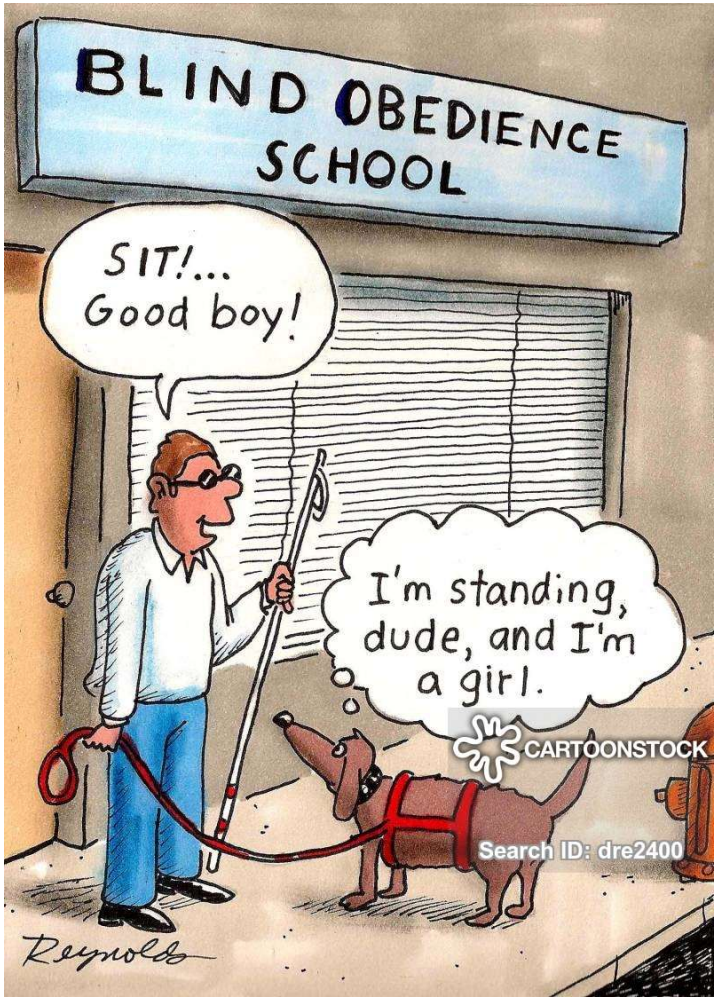
- I can't read your body language
- I'm "flying blind"
- Questions are important and give me feedback
- PLEASE ask questions!



I am working on my mind reading skills but have not perfected them yet.

Your questions and contributions make this class unique and better.

Feedback is a Gift



- Official practice is to "raise your hand".
- But you can unmute, start with "This is _____" and ask or contribute away.
- In the RARE chance it becomes disruptive, I'll announce updated rules.

EDR Students

1	Michael Armstead	ARMSTEAD
2	Steve Arndt	EXPLICO
3	Timothy Austin	WISCONSIN DOT
4	Curtis Beloy	4MOTO
5	Megan Bland	EXPLICO
6	Keith Button	EXPLICO
7	Neal Carter	LUMINOUS
8	Timothy Cheek	DELTA V
9	Eric Edwards	HRYCAY
10	Rosemarie Figueroa	EXPLICO

11	Shaun Jeffs	EXPLICO
12	Ben Merwin	HRYCAY
13	Mark Di Roma	DI ROMA
14	Nathan Rose	LUMINOUS
15	Wendy Sanders	EXPLICO
16	Brett Stoner	FMCSA
17	Ivan Tibavinsky	EXPLICO
18	Adam Tomljenovic	HRYCAY
19	Kevin Toosi	BIOEX
20	Adam Wise	4MOTO

I make a mark by your name each time you ask a question or contribute something.
I will call on the silent ones periodically to encourage questions

Other Housekeeping

- Break times (one long or two shorter?)
- You learn best if you give your full attention to the class. Multi-tasking is very tempting in the virtual world.
- I'll do my best to keep you engaged
- SAE has an end of class learning assessment to get credit for the class. You will get an email when the class is over inviting you to take it.
- (insert directions to bathroom and local restaurants for lunch here…just kidding)

MODULE 1: US EDR STATUS
EDR OVERVIEW

Learning Objectives: Overview

1. Get “big picture” view of EDR’s
2. Basic history of EDR and govt regulation
3. Understand tools to get data
4. Understand typical data gathered and how it has evolved over time
5. Note while minimum data is now standardized, amount of data varies by manufacturer, model, and model year

Some Brief History

1. Early airbag sensors were mechanical, any electronics only ran diagnostics.
2. For 1994 Model Year GM introduced the first electronic accelerometer sensor to measure crash severity, and used it to make airbag deployment decisions..
3. To know if the system performed properly, GM needed to record the inputs and outputs of the system.

Early decisions in EDR's

1. The airbag usually deploys in the first 10-20 milliseconds in a typical 30 mph barrier crash.
2. The average crash pulse lasts 100 to 150 milliseconds.
3. GM measured acceleration 800 times per second.
4. Early control modules had only a very small quantity of EEPROM memory built into the microprocessor.

Early Decisions

1. There was not enough memory to get all the acceleration data.
2. GM decided to calculate the change in velocity from the raw acceleration data and record that once every 10ms.
3. At that time the sensor measured only acceleration in the frontal (longitudinal) direction.
4. GM recorded in airbag deployments, but also decided to record in non-deployments

Early EDR History

1. In 1999 GM attached the airbag control module to the vehicle CAN bus communication system so it could receive information from other modules in the car.
2. GM phased in recording vehicle speed, throttle, brake on/off, and RPM.
3. The purpose of recording was still for GM to know if its airbag system worked correctly.
4. The tool to read this data were proprietary to GM.

Early History Continued

1. The tool to read this data were proprietary to GM.
2. GM asked the supplier of its “Tech 2” dealer diagnostic tool, Vetronix, to make a tool to read crash data for sale to the public.
3. The Vetronix Crash Data Retrieval System was born in 2000.
4. Bosch Diagnostics bought Vetronix in the mid 2000’s.

History Marches On

1. Ford began phasing in electronic sensors in 1997 and recorded limited info.
2. Ford contracted with Vetronix/Bosch to read its limited EDR's in the early 2000's.
3. Chrysler added electronic sensors but did not record anything useful until 2005MY.
4. In 2008 they also chose Bosch CDR to allow their EDR's to be read.
5. Toyota began recording in 2001, and chose Bosch CDR to read its data in 2010

US Government Involvement

1. The US government was reluctant to require EDR's.
2. In 2006 they adopted a regulation that would eventually standardize minimum content of **voluntarily** installed EDR's
3. It also CHANGED THE PURPOSE to be for effective crash reconstruction.
4. The regulation required that data could be accessed by a publicly available tool.
5. Part 563 became effective September 2012.

TABLE I – DATA ELEMENTS REQUIRED FOR ALL VEHICLES EQUIPPED WITH AN EDR

Data Element	Recording Interval/Time[1] (Relative to time zero)	Data Sample Rate Samples per Second
Delta-V, longitudinal	0 to 250 ms	100
Maximum delta-V, longitudinal	0-300 ms	n.a.
Time, maximum delta-V	0-300 ms	n.a.
Speed, vehicle indicated	-5.0 to 0 sec	2
Engine throttle, % full (or accelerator pedal, % full)	-5.0 to 0 sec	2
Service brake, on/off	-5.0 to 0 sec	2
Ignition cycle, crash	-1.0 sec	n.a.
Ignition cycle, download	At time of download	n.a.
Safety belt status, driver	-1.0 sec	n.a.
Frontal air bag warning lamp, on/off	-1.0 sec	n.a.
Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, driver	Event	n.a.
Frontal air bag deployment, time to deploy, in the case of a single stage air bag, or time to first stage deployment, in the case of a multi-stage air bag, right front passenger	Event	n.a.
Multi-event, number of events (1,2)	Event	n.a
Time from event 1 to 2	As needed	n.a.
Complete file recorded (yes, no)	Following other data	n.a.

TABLE II - DATA ELEMENTS REQUIRED FOR VEHICLES UNDER SPECIFIED CONDITIONS

Data Element Name	Condition for Requirement	Recording Interval / Time[1] (Relative to time zero)	Data Sample Rate (Per Second)
Lateral acceleration	If recorded[2]	0-250 ms	500
Longitudinal acceleration	If recorded	0-250 ms	500
Normal acceleration	If recorded	0-250 ms	500
Delta-V, lateral	If recorded	0-250 ms	100
Maximum delta-V, lateral	If recorded	0-300 ms	n.a.
Time maximum delta-V, lateral	If recorded	0-300 ms	n.a.
Time for maximum delta-V, resultant	If recorded	0-300 ms	n.a.
Engine rpm	If recorded	-5.0 to 0 sec	2
Vehicle roll angle	If recorded	-1.0 up to 5.0 sec[3]	10
ABS activity (engaged, non-engaged)	If recorded	-5.0 to 0 sec	2
Stability control (on, off, engaged)	If recorded	-5.0 to 0 sec	2
Steering input	If recorded	-5.0 to 0 sec	2
Safety belt status, right front passenger (buckled, not buckled)	If recorded	-1.0 sec	n.a.
Frontal air bag suppression switch status, right front	If recorded	-1.0 sec	n.a.

Regulations Change Things

1. Most manufacturers that did not yet have an EDR began to install one by the time the regulation came into effect, with notable exceptions of German automakers that prized owner privacy.
2. In 2012 The US government *proposed* requiring EDR's by Sept 2014.
3. The Germans installed EDR's in anticipation of the *proposed* regulation.
4. The proposal was withdrawn in 2019

Introduction - Main Messages

1. EDR's improve recon **accuracy** & get to **causation** – use with **all other** available scene evidence –EDR is also PRIMARY EVIDENCE.
2. EDR strength is speed **PRIOR** TO *BRAKING* OR *LOSS OF CONTROL, &* DRIVER BEHAVIOR on gas and brake pedal. Modern ABS brakes TOOK AWAY your tire marks, only EDR can "give you back" speed prior to braking. Newer data elements give us EVEN MORE INSIGHT into precrash behavior.
3. Event Data Recording capability and data accessibility ***varies widely*** by **manufacturer, model *and* model year.** Bosch Crash Data Retrieval system has widest coverage and acceptance, but some EDR's require other special tools or procedures get EDR data

Main Messages

4. NHTSA Part 563 EDR rule Sept 2012 was a game changer – required **Minimum** Data in US vehicles equipped with EDR
5. EDR not required but 99% of US 2020 vehicles have one.
6. EDR's will NEVER replace human beings in figuring out what really happened.
7. US Driver Privacy Act Dec 2015 requires owner permission or a warrant to access.
8. AEB (auto braking) + other driver assist tech= fewer crashes + more data
9. Software & Hardware to access change constantly, keep up to date!

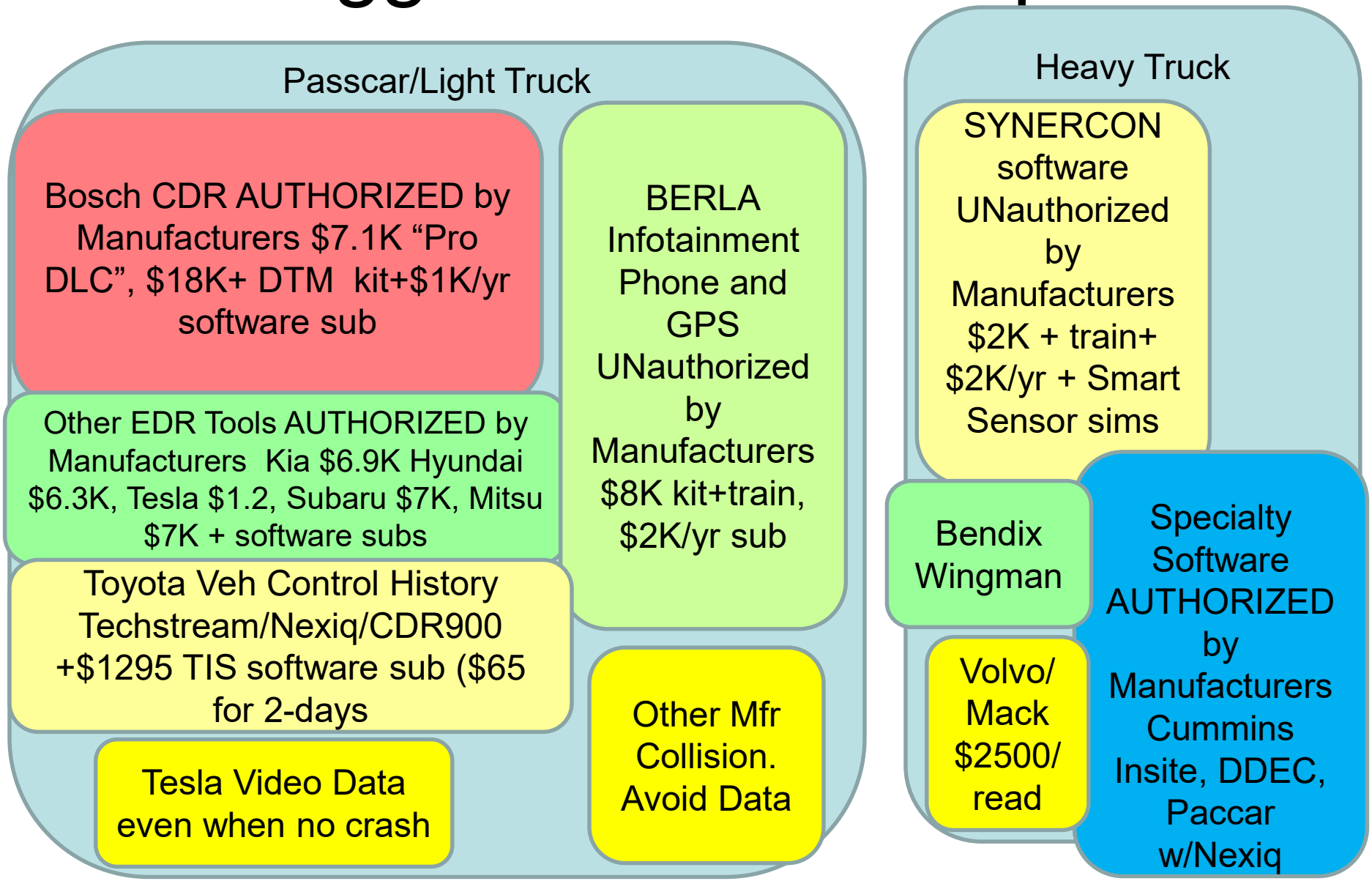
Keeping EDR's in Perspective

1. There are 305 million vehicles registered in the US. 63% have an EDR with data accessible by the Bosch CDR system
2. 87+% of 2020's are supported by Bosch.
3. Another 12% have EDR accessible by other tools. **99%** have EDR.
4. With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles even if only one has a 563 EDR.

Worldwide EDR installation significantly lags US

1. **Only a few mfrs voluntarily install EDR worldwide**
2. **New EU reg 3/2022** new design, 3/2024 old
3. Europe and Australia – mostly Toyota, Volvo, FCA, Holden, 2018+ VW.
4. New Korean reg Dec 2015.
5. Japan 563 type reg, Tokyo police 50 kits.
6. China making EDR reg now for 2021/23.

The bigger vehicle data picture



Bosch Crash Data Retrieval Kit

1. Most widely used tool in the US and worldwide. Reads 63% of US vehicles.
2. 19 Manufacturers confidentially give Bosch how to access EDR memory and interpret it
3. Program on laptop talks to interface module which communicates to car thru DLC if vehicle electrical system still functions
4. MANY direct to module cables in case vehicle electrical system does not work!

Bosch CDR Kit

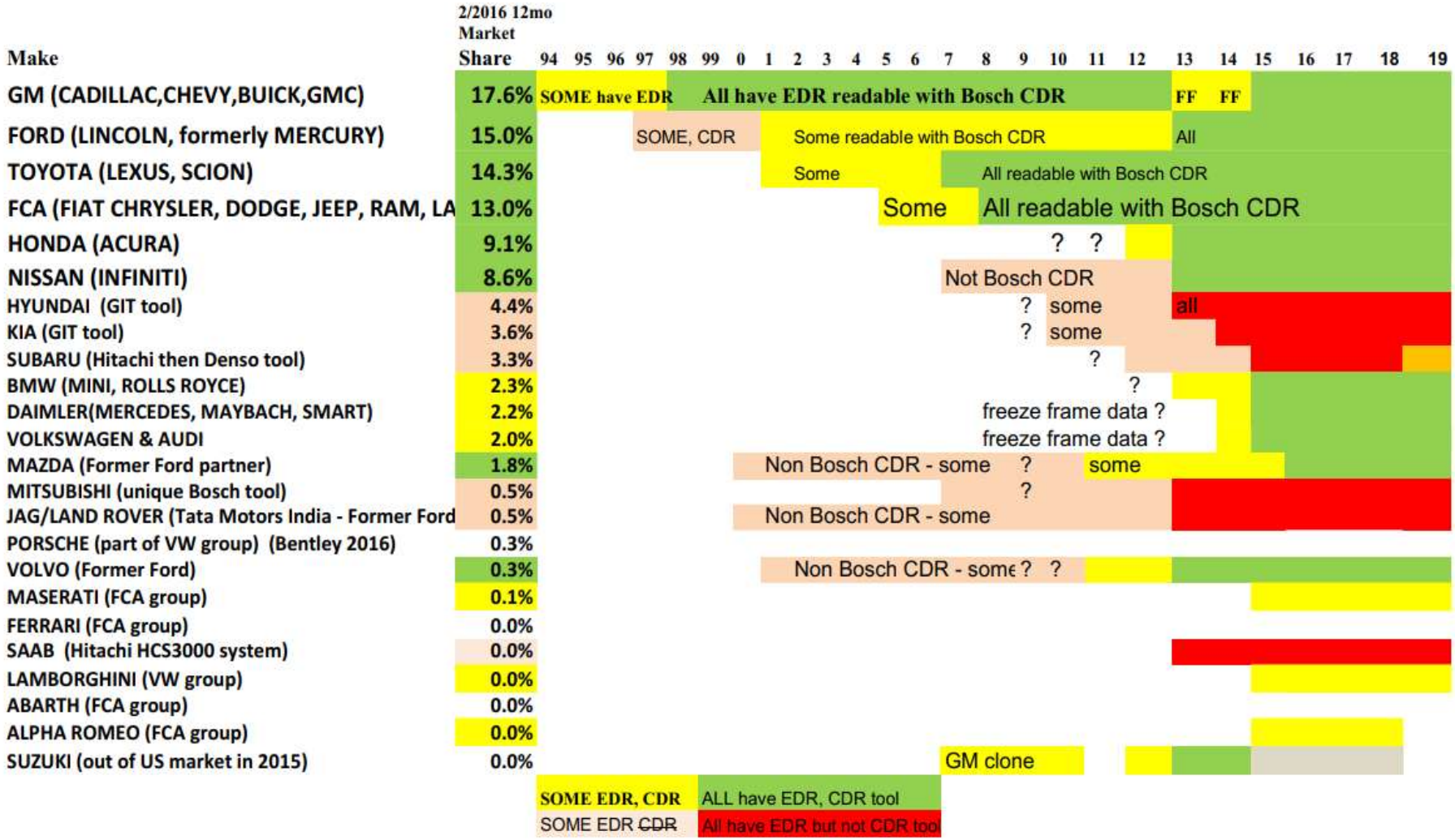
5. Reads data 3 times and compares reads to make sure they are identical
6. Encrypts data in raw data file to reduce risk of data tampering
7. Enables raw data file to be reinterpreted in latest software in case bugs are found
8. Provides easy to read PDF reports
9. Widely accepted by govt, safety research, accident reconstruction, law enforcement, insurance fraud investigators, and OEM's.

US Lt. Veh EDR & Tool Used

Manufacturer	Annual Sales 4Q 18 to 3Q 19	Percent	EDR First Known Use	Tool?	
GM (Chevy, Buick, GMC, Cadillac)	2,920,580	17.0%	1994	Bosch CDR	
Ford (Ford, Lincoln)	2,404,373	14.0%	1997	Bosch CDR 2001+	
Toyota (includes Lexus, Scion)	2,381,734	13.9%	2001	Bosch CDR	
Fiat Chrysler (includes Dodge, Ram, Jeep, Lancia	2,216,366	12.9%	2005	Bosch CDR	
Honda (includes Acura)	1,604,240	9.4%	2012	Bosch CDR 2012+	
Nissan (includes Infinity, Renault, EXCLUDES Mitsubishi)	1,413,835	8.3%	2006	Nissan Consult	CDR 2013+
Hyundai (Includes Kia, Genesis)	1,309,862	7.6%	2010	GIT tool co.	
Subaru	702,046	4.1%	2012	Denso +SSM	CDR 2019
BMW (incudes Mini, Rolls Royce)	640,003	3.7%	2013	Bosch CDR	
Volkswagen (Includes Audi, Porsche, Bentley)	355,692	2.1%	2014	Bosch CDR	
Daimler Benz (Mercedes, Smart	348,173	2.0%	2014	Bosch CDR	
Mazda	273,368	1.6%	2011	Bosch CDR 2011+	
Tesla	216,125	1.3%	2012	Tesla Tool	
Tata (Jaguar, Land Rover)	125,388	0.7%		Bosch SPX	
Mitsubishi (Part of Nissan Renault Alliance)	120,000	0.7%	2006	Bosch SPX	CDR 2019
Volvo	102,059	0.6%	early	Bosch CDR 2011+	
Industry Total	17,133,844.00	100.0%			

EDR Phase-in by Mfr

Data is incomplete for smaller mfrs with no public tool



2020 Recording Capability

EDR Major Data Elements by Manufacturer (All Mfrs with EDR have the required 15 data elements, all are not listed)

Information is taken from Bosch CDR sample files, Ruth cases, and 2012 MY practice projected into 2019 model year

	Part 563 Table 1								Part 563 Table 2						Beyond 563									
Mfr	Event Complete	Key Cycles	Long. ΔV	Speed/Brake/ Throttle or Accel	Multi Event	Lateral ΔV	RPM	Steering Angle	ACM Acceleration	ABS on/off	Roll Angle	ESC Long Accel	ESC Lat Accel	Yaw Rate	Tire Pressure	Panic Brake Assist	Wheel Speeds	Roll Rate	Brake Pressure	OTHER
GM	X	X	X	5@0.5 TA	X	X	X	-	X, Y, Z 300@2	X		-	in roll		LO	-	-	1 @ 10ms		2016 ASCM w supercruise
Ford	X	X	X	5@0.5 A	X	X	X	5@0.1	-	X		5@0.1		5@0.1	LO		-	5@0.1	**	2017 AB12 precrash 0.2 sec
Chrysler	X	X	X	5@0.1 TA	X	X	X	5@0.1	-	X		-		5@0.1	PSI	X	X		2018	ACC set speed 2018, Avoidance
Toyota	X	X	X	5@0.5 TA	X	Side	X	5@0.5	-		Peak&atTRG	5	in roll	5@0.5	-	-	-	In Roll	5@0.5	Ped 2018, Veh history rpt
Honda	X	X	X	5@0.5 TA	X	X	X	5@0.5	X, Y, Z 250@10	X	-1+5@0.1	-		-	-	-	-			AEmBrk. Lanekeep
Nissan	X	X	X	5@0.5 A	X	X	X	5@0.5	X, Y 250@10	-	-	-		-	-	-	?			
Mazda	X	X	X	5@0.5 T	X	X	X		X, Y, Z 250@10	-							-			
Volvo	X	X	X	5@0.5T	X	2016 XC90	-	2016 XC90	X,Y,Z 250@10	2016 XC90	-1+0.3@0.1									
BMW	X	X	X	5@0.5A	X		X	5@0.5	X, Y 300@10ms	X	-									
Mercedes	X	X	X	5@0.5A	X	2017 E class adds														
VW	X	X	X	5@0.5A	X	X														Time of Crash
Hyundai/Kia	X	X	X	5@0.5	X	X	X	5@0.5	X,Y,Z 250@10	X	-1+5@0.1	-		-	-	-	-		19 Forte	
Subaru	X	X	X	5@0.5A	X	X	X	5@0.5		X								-1 to+1.5		Eyesight
Mitsubishi	X	X	X	5@0.5A	X	X	X	5@0.5	X, Y 250@10		-1+5@0.1									

Bosch CDR Hardware Update

1. *New interface module CDR-900 kit with new communication protocols needed in newer vehicles (*CAN-HD and Ethernet) Bigger 19-pin connector needed (vs 15).
2. *New cables are being made with 19 pins, you will NEED a CDR 900 to read new manufacturers and models.*
3. *The CDR900 is a derivative of a proven auto industry Bosch scan tool*
4. *CDR 900 must be registered to your laptop (network device) before use.*



CDR 900 Upgrade Kit

$4000

CDR900

1. Initially for NEW vehicles
2. Expected to handle <u>most</u> legacy vehicles eventually

.

CDR900 Upgrade Kit



During Changeover

- Bosch is working to support “Legacy” vehicles with CDR-900.
- Legacy coverage will take more than 1.5 years to implement, test, and release – so the CAN+ interface may still be needed through 2021 (or longer???) .
- The help file clearly identifies which interface(s) will work with each vehicle
- CDR-900 comes with adapter to old 15-pin cables, they will still work.

New users need **both** CAN+ and CDR900



MSRP: $7300

plus shipping & handling

Add to Cart ›

Bosch CDR Pro Tool Kit

The CDR Pro Tool Kit is the heart and soul of the Bosch CDR Tool. This package contains all of the hardware required to perform a DLC/OBD retrieval of EDR data in Bosch supported vehicles. It includes both the CDR 900 and CANplus vehicle interface modules. This package also comes with a 1-year CDR software subscription. NOTE: this package does NOT contain any hardware to perform a "direct-to-module (D2M)" retrieval of EDR data. Please see the accessory cable/adapter packages below for D2M options.

- CDR 900 vehicle interface module
- CDR 900 power cable
- CDR 900/CDR CAN*plus* power supply cable
- CDR 900/CDR CAN*plus* CDR cigarette power adapter
- CDR 900 DLC/OBD cable
- CDR 900 wireless dongles (2)
- CDR 900 extension cable
- CDR 900 legacy cable adapter
- CDR 900 USB cable
- CDR CAN*plus* vehicle interface module
- CDR CAN*plus* DLC/OBD cable
- CDR CAN*plus* GM DLC/OBD cable
- CDR CAN*plus* DLC/OBD adapters (3)
- CDR CAN*plus* PCM adapter
- CDR CAN*plus* USB to Serial port cable
- Deluxe repowering rig
- 1-year CDR software subscription
- Custom carrying case
- Coupon for **FREE online class**: How to Use the Bosch CDR Tool

CDR 900 details

1. CDR900 integrates power management (no external relays/adapters).
2. CDR900 uses the existing 2.5 amp current limited power supply, but new ACM's may drive bigger supply later.
3. Includes WIRELESS laptop to interface
4. CDR900 does NOT integrate the current CDR500 Flex Ray adapter – still needed!
5. CDR 900 firmware must be flashed initially

CDR900 Tips

1. CDR900 is a network device, VPN's do NOT LIKE other network devices, they are designed to keep them out. CDR900 does not like laptops with firewalls and other virus protection and lots of other programs – suggest using a *dedicated laptop* with few other programs installed.
2. VCI Manager and CDR are *two different programs*, you MUST close VCI manager before CDR program will recognize CDR900!

HARDWARE CHECKLIST 19.3

Part Number	Description	Released
1699200596	Cable, ID828 CDR Nissan ACM	17.5
1699200618	Cable, ID829 CDR FCA ACM	17.5
1699200553	Cable, ID826 CDR Audi ACM (requires CDR 500 Adapter)	17.6
1699200646	Cable, ID830 CDR Volvo ACM (requires CDR 500 Adapter)	17.7
1699200630	Kit, CDR 900 Upgrade, with Wireless	17.8
1699200696	Cable, ID831 CDR Mazda ACM	17.8
1699200697	Cable, ID832 CDR Daimler ACM	17.8
1699200778	Cable, ID835, CDR Subaru ACM 1	19.0
1699200779	Cable, ID836, CDR Subaru ACM 2	19.0
1699200783	Cable, ID837, CDR Mitsubishi ACM 1	19.0
1699200784	Cable, ID838, CDR Mitsubishi ACM 2	19.0
1699200785	Cable, ID839, CDR Mitsubishi ACM 3	19.0
1699200786	Cable, ID840, CDR Mitsubishi ACM 4	19.0
1699200776	Cable, ID833, CDR GM ASCM 1	19.1
1699200777	Cable, ID834, CDR GM ASCM 2	19.1
1699200797	Cable, ID842, CDR VW Group ACM 4	19.1
1699200800	Cable, ID843, CDR Mazda ACM	19.1
1699200838	Cable, ID845, CDR GM ACM	19.1

New 3-digit Part ID

Gen 4 Pop Up Hood

2019

software to use them not out yet

New Back Powering Gear Needed

1. Manufacturers are replacing mini and micro fuses with Micro 2 and Micro 3 fuses to pack more fuses into smaller spaces. The complete back power kit must now have several different yellow clips to fit all the different vehicles on the road. GM started it, Ford followed, more to come. Mini fits into sub mini, don't need sub mini



Buy General Technologies Corp GTC063 set of 6 (including 2 bigger ones you don't really need) at the site below for $19.95

https://www.amazon.com/General-Technologies-Corp-GTC063-Connector/dp/B0787JKMTH?ref_=bl_dp_s_web_9003703011

EMERGING TRENDS

1. More manufacturers are becoming concerned about cyber security, of hackers trying to take over vehicle control systems.
2. Many are installing “gate-way” modules or features to protect ECU’s from hacking.
3. For CDR DLC users, “back powering” may have to power ACM fuse AND cyber security (gateway) module fuse. Of course Direct to module is still an alternative.
4. Recently needed on 2017 RAM 3500, 2018 Toyota

Recent CDR Software Updates

- 17.8 – July 31, 2018 – **CDR900** + more 19's
- 17.9 – Sept 18, 2018 – DLC GM ASCM
- 17.10 Dec. 20, 2018 – 2018-19 VW/Audi Worldwide, more 2019's, some GM 2020's
- 18.0 Feb 20, 2019 **Subaru** 19 Forstr/Acsendr
- 19.0 June 6, 2019 **Mitsubishi** Outlanders
- 19.1 Sept. 20, 2019, 15+ Subaru Outback, GM ASCM reports, more 2020's
- 19.2 Dec. 6 Legacy GM ASCM's, + 2020's
- 19.3 –Dec.23 – CDR900 Toyota support
- 19.3.1 Feb 13, 2020 removed 30 sec delay

Bosch CDR Recent News

1. New CDR 900 interface – old interface lasted 13 years – technology has changed!
2. New GM active safety control module gives access to semi-autonomous vehicle data
3. VW/Audi going worldwide with EDR
4. The covered vehicles on the road with CDR is 63% and climbing about 3%/yr. The total available including mfrs with non-CDR tools is much more!!
5. Subaru new models in CDR, legacy to follow, Mitsubishi added June 2019.

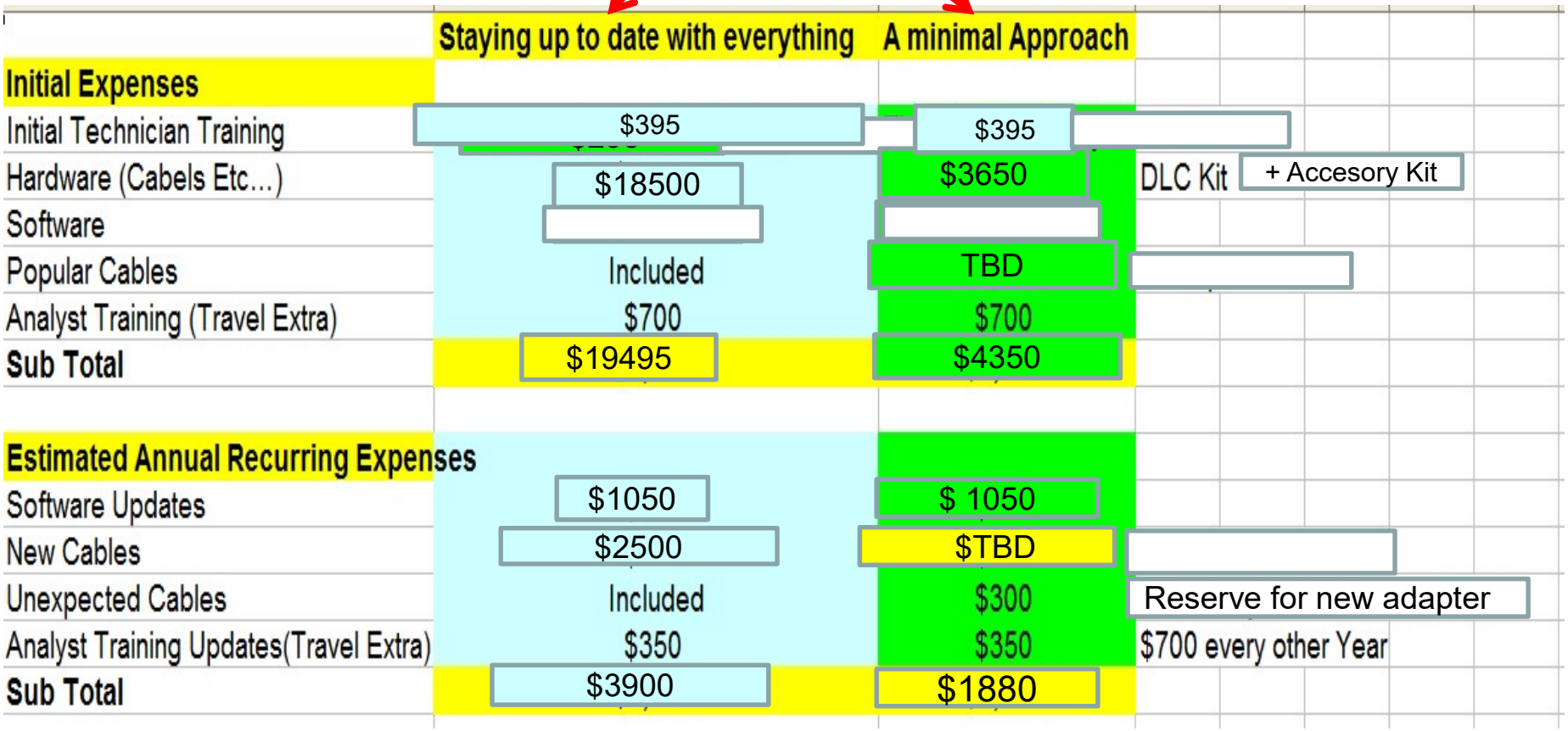
What's Coming?

1. 8 new DTM cables (budget $2,000) – Subaru 2020 Legacy/Outback (June), Honda, Daimler.
2. Back model Subaru coverage expected in 19.4 "soon" – end April 2020
3. **New Toyota EDR (Cable 827) in 19.4**
4. **CDR900 support - Honda, Nissan, Mazda**
5. **2020 Porsche Taycan coverage (and other newer Porsche models with time)**
6. **More GM ASCM coverage (EOCM3)**

CDR Budgeting

Two approaches to a budget

Add one time new CDR900
$3,800



	Staying up to date with everything	A minimal Approach	
Initial Expenses			
Initial Technician Training	$395	$395	
Hardware (Cabels Etc...)	$18500	$3650	DLC Kit + Accesory Kit
Software			
Popular Cables	Included	TBD	
Analyst Training (Travel Extra)	$700	$700	
Sub Total	$19495	$4350	
Estimated Annual Recurring Expenses			
Software Updates	$1050	$ 1050	
New Cables	$2500	$TBD	
Unexpected Cables	Included	$300	Reserve for new adapter
Analyst Training Updates(Travel Extra)	$350	$350	$700 every other Year
Sub Total	$3900	$1880	

New Tool for Insurance?

1. Collision Sciences based in Toronto has developed a DLC only tool and Bluetooth software that runs from your phone to read EDR data from vehicles with functioning electrical systems. Intended customer is insurance.
2. Entry price for the hardware is only $150
3. Raw data is uploaded to cloud, free teaser preview
4. Full reports cost $300 per readout and include value added analysis and risk of fraud and injury assessment.
5. Developers are talented hackers - they are NOT getting decoding data from manufacturers – reverse engineering
6. Works on Kia/Hyundai and most CDR supported vehicles
7. Unclear if Bosch or GIT could claim software license violation and go after them – coverage mirrors CDR/GIT

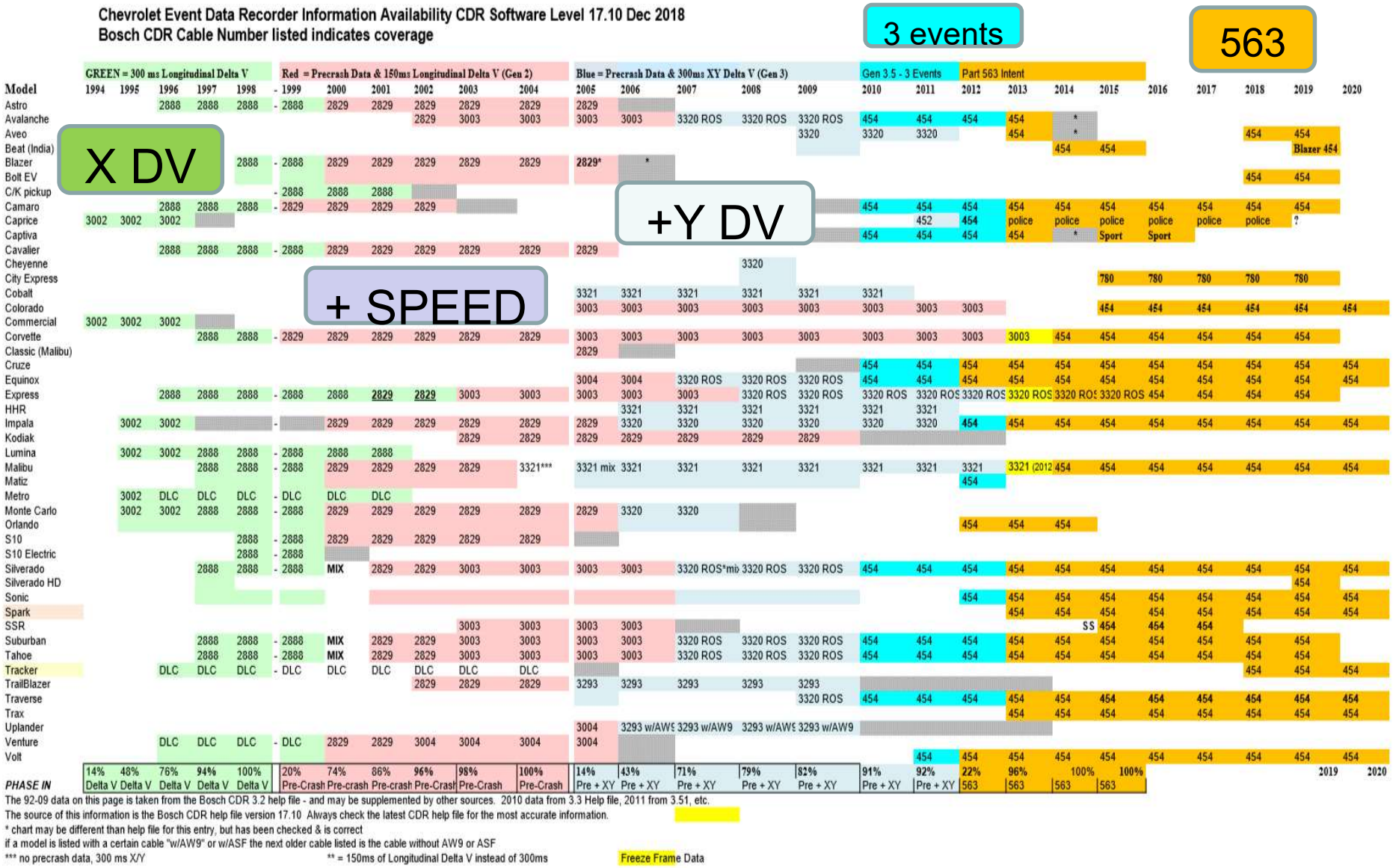
NHTSA NASS DATABASE

1. https://www.nhtsa.gov/research-data/national-automotive-sampling-system-nass
2. NASS contains 13,898 EDR reports.
3. For privacy reasons gov't had to redact some information, so copies are PDF'd – the ".cdrx" raw data file is not available for re-interpretation in latest software
4. NASS ended in 2015, is replaced by CISS in 2016

NHTSA CISS DATABASE

1. https://crashviewer.nhtsa.dot.gov/CISS/SearchIndex
2. In 2018 NHTSA loaded 125 2016 Crash Investigation Sampling System cases. 2,068 2017 cases were uploaded late Sept, 2019. 2018 cases expected 6/2020.
3. Searchable by manufacturer, model year, model, EDR data available, driver assist technology available, etc
4. Great source of info on what to expect in your vehicle's EDR, confirmation of EDR polarity, etc.

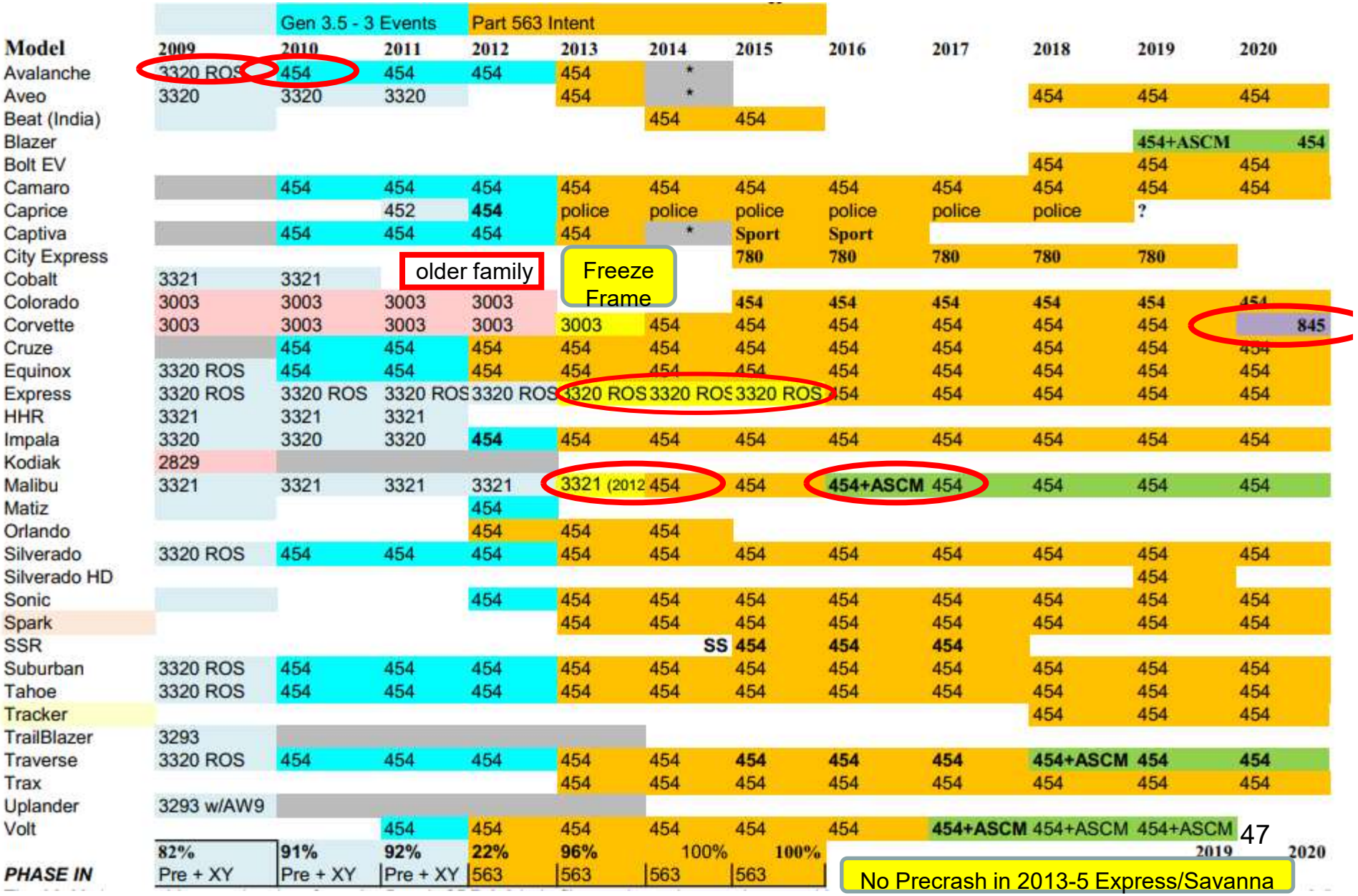
Chevrolet SDM Phase In 87%



GM full chart printed in color in your book and is on student reference disk

Chevrolet Recent Changes 87% 563

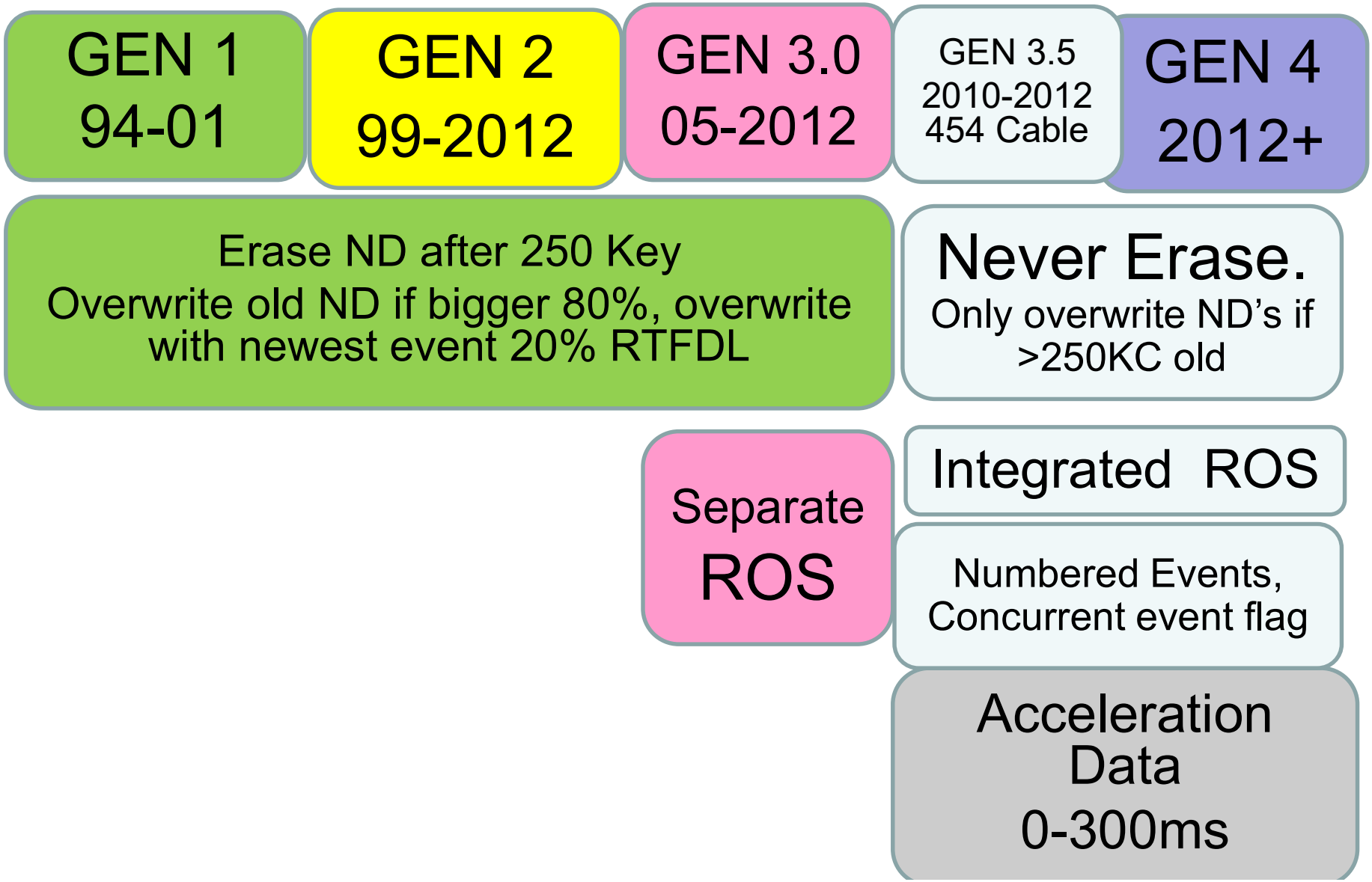
1 animation



GM EDR Evolution

GEN 1 94-01	GEN 2 99-2012	GEN 3.0 05-2012	GEN 3.5 2010-2012 454 Cable	GEN 4 2012+
No Precrash	Precrash 5 @ 1 sec for BOTH events	Precrash 5 @ 0.5 or 1 sec ONE SET of precrash	Precrash 5 @ 0.5 or 1 sec For each event	Precrash 10 @ 0.5 sec
2 EVENTS - FIRST D, One ND/ DLE, replace SDM after D			3 Events ND's or D's, replace after 3D's	
Record on frontal airbag algo wakeup		Record on 5mph X or Y DV,		
300ms XDV from wakeup, must be neg	150ms XDV, must be neg, 50ms before D 100 after	300ms X+Y DV, Neg or Pos, some D's -70 to +220, some 300ms after wakeup		

GM EDR Evolution #2



GM **Gen 2** (RED Group)

Typical Pre-Crash Data

Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle
-5	96	2816	84
-4	97	3712	100
-3	93	3776	100
-2	56	1920	0
-1	13	640	0

Seconds Before AE	Brake Switch Circuit Status
-8	OFF
-7	OFF
-6	OFF
-5	OFF
-4	OFF
-3	OFF
-2	ON
-1	ON

Gold “Group 4” 2012+ Part 563 Intent

Pre-Crash Data -5.0 to -0.5 sec (Event Record 1)

Times (sec)	Accelerator Pedal, % Full (Accelerator Pedal Position)	Service Brake (Brake Switch Circuit State)	Engine RPM (Engine Speed)	Engine Throttle, % Full (Throttle Position)	Speed, Vehicle Indicated (Vehicle Speed) (MPH [km/h])
-5.0	0	On	640	2	0 [0]
-4.5	0	On	640	3	0 [0]
-4.0	0	Off	640	3	0 [0]
-3.5	21	Off	704	16	1 [1]
-3.0	21	Off	1408	20	1 [2]
-2.5	23	Off	1600	23	3 [5]
-2.0	25	Off	1856	25	6 [9]
-1.5	25	Off	2048	25	8 [13]
-1.0	0	Off	2048	21	10 [16]
-0.5	0	Off	1856	5	11 [17]

10 Samples

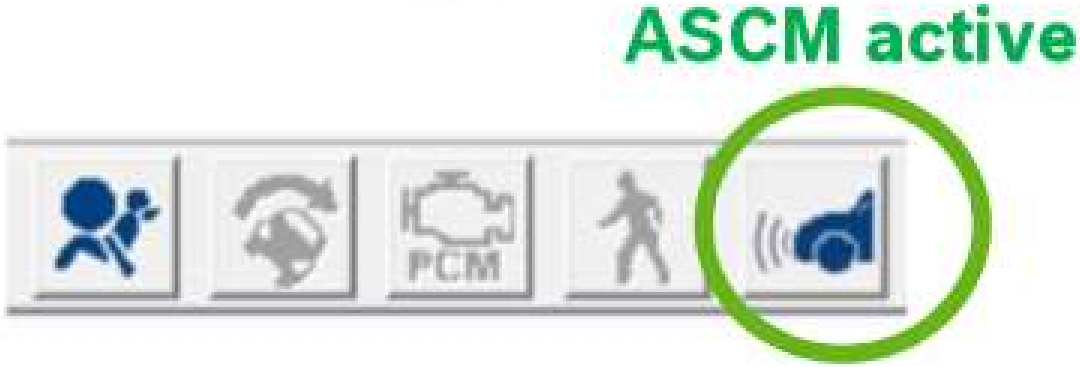
GM Auto Emergency Braking

1. AEB will prevent many crashes and reduce the severity of crashes still happening
2. If there is still a crash, the AEB system may store up to 3 camera images:
   a) 4 sec before event
   b) At event
   c) 4 sec after event
3. Not accessible with CDR – must be requested with proper legal authority from GM

GM Active Safety Control Module

1. First installed in 2013 Cadillac ATS
2. Only with GM option UGN or UKL
3. Initially released in CDR 17.9 to get raw data file only for future interpretation.
4. Interpretation capability added in 19.1
5. REQUIRES CDR 900 interface!
6. Some vehicles have TWO ASCM modules.

GM Active Safety Module



1. Must be downloaded separately from ACM
2. Icon will appear after VIN input for any vehicle that MIGHT have Super Cruise
3. Module /data will only be present if option “UKL” (Super Cruise) or “UGN” (Adv Driver Assist) has been purchased for the subject vehicle.

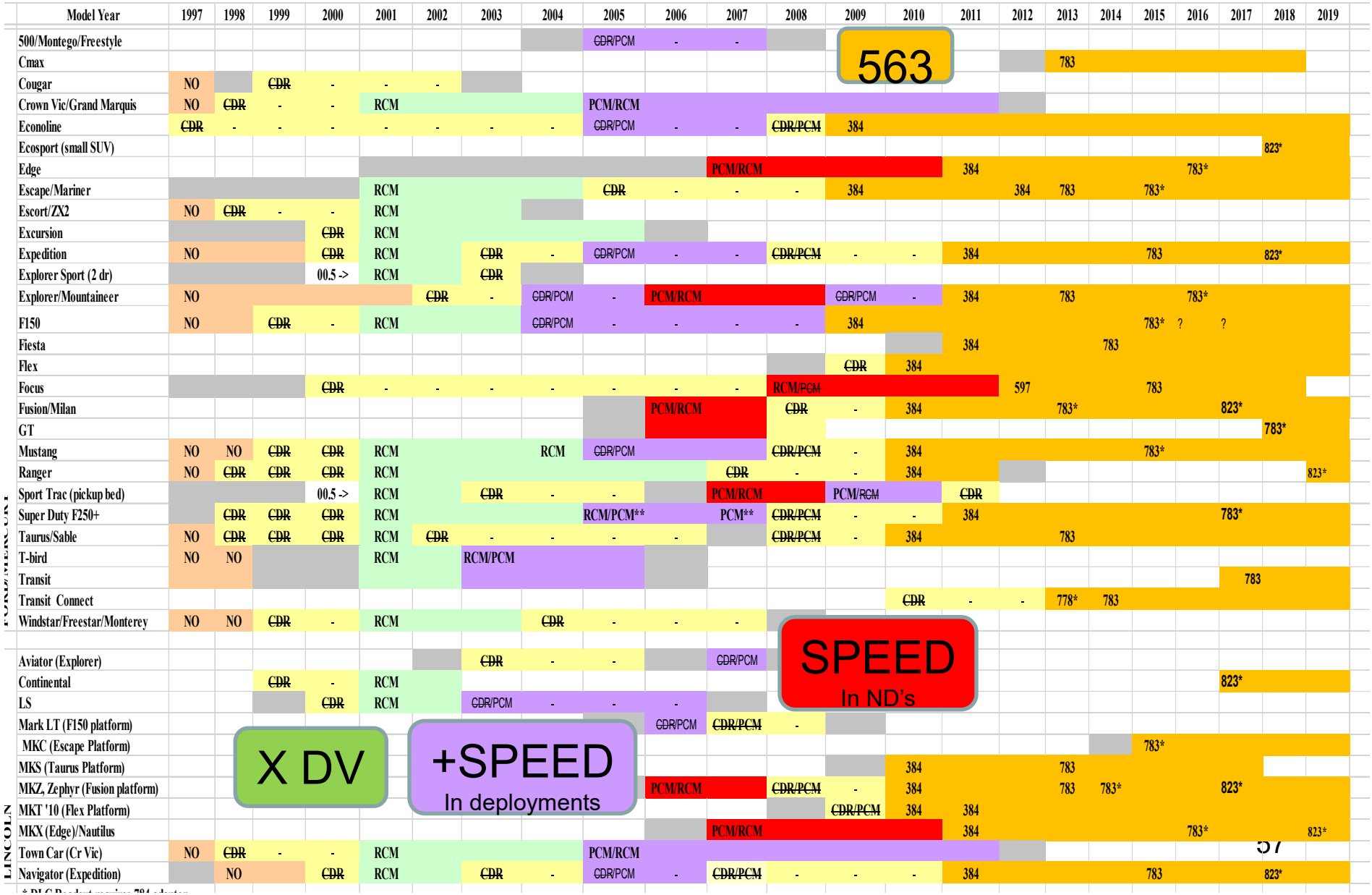
GM ASCM CDR 19.2 Coverage
ASCM = Advanced Safety Control Module

Model	Location	Model Year							
3 digits Bosch CDR cable ID shown		2013	2014	2015	2016	2017	2018	2019	2020
BUICK									
ENCLAVE	cargo left front of wheel well						833		
ENVISION	rear seat left trim by wheel wl				833				
LACROSSE	trunk left					833			
REGAL	trunk left								
CADILLAC									
CT4	? Not in help file 19.2 yet								833
CT6 w/o UKL	Trunk right					833	without supercruise		
CT6 w/UKL	Primary right, secondary left				834		Supercruise Option Code UKL		
CTS SEDAN	trunk left		833						
CTS	trunk left				833				
ESCALADE	under left qtr window trim			833					
SRX	under left rear seat	833							
XT4	cargo left front of wheel well							833	
XT5	cargo left front of wheel well					833			
CHEVROLET									
BLAZER	trunk left							833	
MALIBU	trunk left				833				
TRAVERSE	trunk right						833		
VOLT	trunk left					833			
GMC									
ACADIA	cargo left front of wheel well					833			

GM Active Safety Module

1. Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
2. Data is recorded in non-volatile memory at the next proper power-down of the module. Data may **not** be stored if the battery power is lost before the power-down.

Ford CDR 17.8 Coverage 69%



	Model Year	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
FORD/MERCURY	500/Montego/Freestyle									~~CDR~~/PCM	-	-												
	Cmax																	783						
	Cougar	NO		~~CDR~~	-	-	-																	
	Crown Vic/Grand Marquis	NO	~~CDR~~	-	-	RCM				PCM/RCM														
	Econoline	~~CDR~~	-	-	-	-	-	-	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	384										
	Ecosport (small SUV)																						823*	
	Edge											PCM/RCM				384					783*			
	Escape/Mariner					RCM				~~CDR~~	-	-	-	384			384	783		783*				
	Escort/ZX2	NO	~~CDR~~	-	-	RCM																		
	Excursion				~~CDR~~	RCM																		
	Expedition	NO			~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	-	~~CDR/PCM~~	-	-	384				783			823*	
	Explorer Sport (2 dr)				00.5 ->	RCM		~~CDR~~																
	Explorer/Mountaineer	NO					~~CDR~~	-	~~CDR~~/PCM	-	PCM/RCM			~~CDR~~/PCM	-	384		783			783*			
	F150	NO		~~CDR~~	-	RCM			~~CDR~~/PCM	-	-	-	-	384						783*	?	?			
	Fiesta															384			783					
	Flex													~~CDR~~	384									
	Focus				~~CDR~~	-	-	-	-	-	-	-	RCM/~~PCM~~				597			783				
	Fusion/Milan										PCM/RCM		~~CDR~~	-	384			783*				823*		
	GT																						783*	
	Mustang	NO	NO	~~CDR~~	~~CDR~~	RCM			RCM	~~CDR~~/PCM			~~CDR/PCM~~	-	384					783*				
	Ranger	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM						~~CDR~~	-	-	384									823*
	Sport Trac (pickup bed)				00.5 ->	RCM		~~CDR~~	-	-		PCM/RCM		PCM/~~RCM~~		~~CDR~~								
	Super Duty F250+		~~CDR~~	~~CDR~~	~~CDR~~	RCM				RCM/PCM**		PCM**	~~CDR/PCM~~	-	-	384						783*		
	Taurus/Sable	NO	~~CDR~~	~~CDR~~	~~CDR~~	RCM	~~CDR~~	-	-	-	-		~~CDR/PCM~~	-	384			783						
	T-bird	NO	NO			RCM		RCM/PCM																
	Transit																					783		
	Transit Connect														~~CDR~~	-	-	778*	783					
	Windstar/Freestar/Monterey	NO	NO	~~CDR~~	-	RCM			~~CDR~~	-	-	-												
LINCOLN	Aviator (Explorer)							~~CDR~~	-	-		~~CDR~~/PCM												
	Continental			~~CDR~~	-	RCM																823*		
	LS				~~CDR~~	RCM		~~CDR~~/PCM	-	-	-													
	Mark LT (F150 platform)										~~CDR~~/PCM	~~CDR/PCM~~	-											
	MKC (Escape Platform)																			783*				
	MKS (Taurus Platform)														384			783						
	MKZ, Zephyr (Fusion platform)										PCM/RCM		~~CDR/PCM~~	-	384			783	783*			823*		
	MKT '10 (Flex Platform)													~~CDR/PCM~~	384	384								
	MKX (Edge)/Nautilus											PCM/RCM				384					783*			823*
	Town Car (Cr Vic)	NO	~~CDR~~	-	-	RCM				PCM/RCM														
	Navigator (Expedition)		NO		~~CDR~~	RCM		~~CDR~~	-	~~CDR~~/PCM	-	~~CDR/PCM~~	-	-	-	384				783			823*	

Ford Gold group **Pre- Crash** Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (First Record)

Times (sec)	Speed vehicle indicated MPH [km/h]	Accelerator pedal, % full	Service brake, on/off	ABS activity (engaged, non-engaged)	Stability control (engaged, non-engaged)
- 5.0	50.9 [82.0]	0.0	Off	non-engaged	non-engaged
- 4.5	50.6 [81.5]	0.0	Off	non-engaged	non-engaged
- 4.0	50.2 [80.7]	0.0	Off	non-engaged	non-engaged
- 3.5	49.7 [80.1]	0.0	Off	non-engaged	non-engaged
- 3.0	49.3 [79.3]	0.0	Off	non-engaged	non-engaged
- 2.5	48.8 [78.6]	0.0	Off	non-engaged	non-engaged
- 2.0	48.6 [78.3]	0.0	Off	non-engaged	non-engaged
- 1.5	48.3 [77.8]	7.5	Off	non-engaged	non-engaged
- 1.0	48.2 [77.5]	0.0	Off	non-engaged	non-engaged
- 0.5	47.7 [76.8]	0.0	Off	non-engaged	non-engaged
0.0	32.7 [52.7]	0.0	On	engaged	non-engaged

Ford w/Stability Control Data 19.3

Model Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
CMAX					783							
ECONOLINE	384 RC6											
ECOSPORT										823*		
EDGE			384					783*				823*
ESCAPE	384			384	783		783*					823*
EXPEDITION	-	-	384				783			823*		
EXPLORER		-	384		783	RC7		783*				823*
F150	384						783*			RC7P		
FIESTA			384			783						
FLEX		384										
FOCUS				597 Bosch AB10			783					
FUSION		384			783*			AB10P	823*			
GT										783*		
MUSTANG		384					783*				RC7P	
RANGER		384									823*	
SUPER DUTY F250-550		-	384						783* RC7P			
TAURUS/SABLE		384			783 RC7							
TRANSIT (big van)									783			
TRANSIT CONNECT (small)					778*	783						
AVIATOR (Explorer platform)												823*
Continental									823*			
MKC/CORSAIR (Escape Platform)							783*					823*
MKS (Taurus Platform)		384			783							
MKZ, Zephyr (Fusion platform)		384			783	783*			823* AB12			
MKT '10 (Flex Platform)		384	384									
MKX/NAUTILUS (Edge platform)			384					783*			823*	
NAVIGATOR (Expedition platform)			384				783			823*		

Ford Part 563 EDR Subfamilies

Autoliv RC6
2009 Escape
384 Cable
+RC6_2011
(RC6X)

Bosch AB10
2012 Focus
597 Cable

Autoliv RC7
2013 Taurus
783 Cable
+RC7P upgrade

Bosch
AB10P
2015
Focus
783 Cable

RC7P

Bosch AB12
2017 Fusion
823 Cable

Precrash 5 sec @0.5 sec intervals, 250msec X&Y DV

2nd DV gra ph

Precrash
5 sec
@ *0.2* sec

Stability Control Sys Data 5 sec @0.1 sec intervals

RC6 last speed data
0.5 sec plus delay
RC6_2011 0.5 sec

Last data point within 0.5 sec

?

?

Within
0.2
sec

Incomplete Data Limitations

Fairly Complete Data Limitations

Available on my website

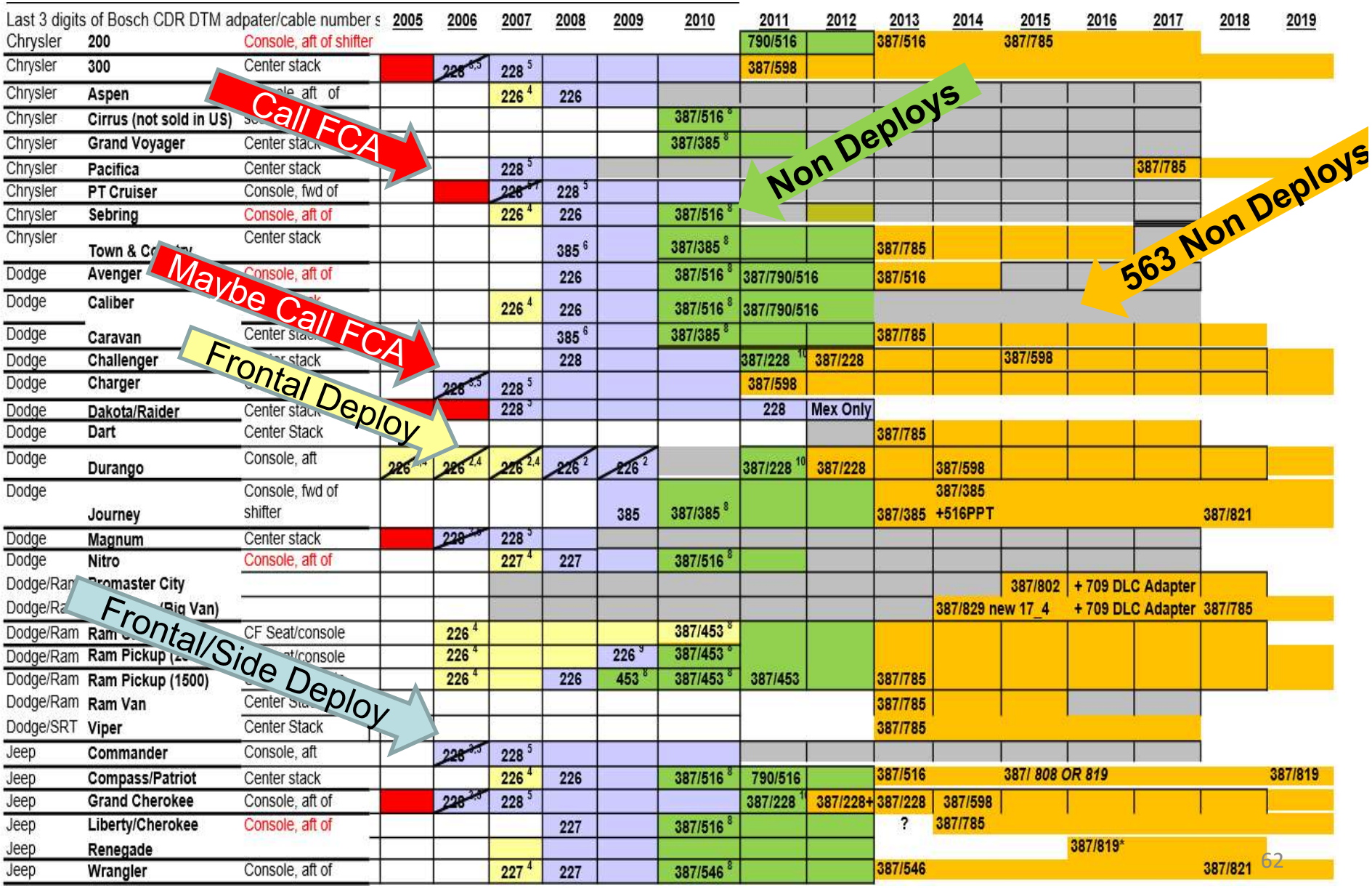
Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 11-24-2019, after the original presentation at the Jan 2016 CDR Summit)

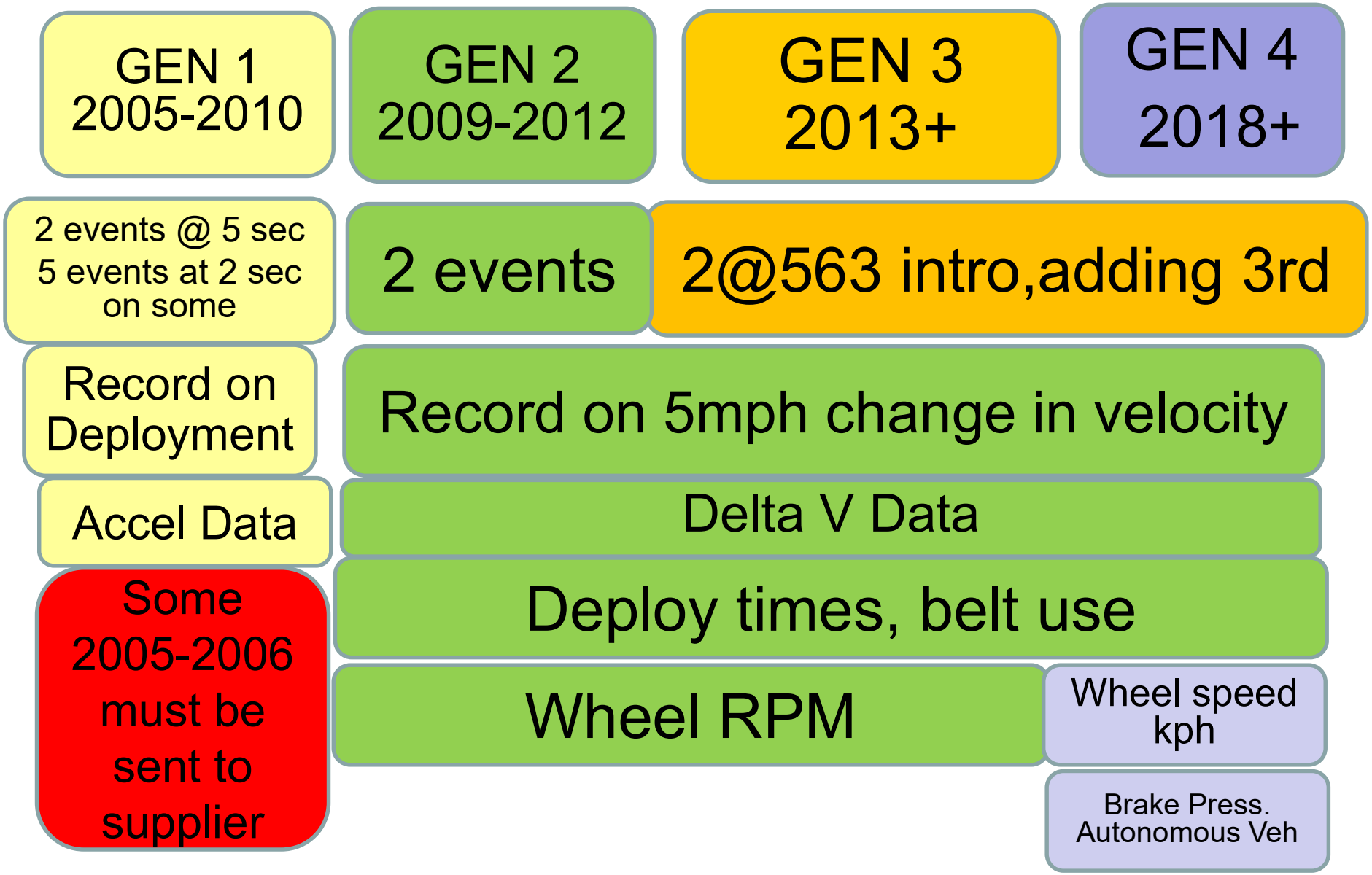


FCA 19.3 CDR Coverage 64%



Last 3 digits of Bosch CDR DTM adpater/cable number s			2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Chrysler	200	Console, aft of shifter							790/516		387/516		387/785				
Chrysler	300	Center stack		~~226~~ [3,5]	228 [5]				387/598								
Chrysler	Aspen	...le, aft of			226 [4]	226											
Chrysler	Cirrus (not sold in US)							387/516 [8]									
Chrysler	Grand Voyager	Center stack						387/385 [8]									
Chrysler	Pacifica	Center stack			228 [5]										387/785		
Chrysler	PT Cruiser	Console, fwd of			~~226~~ [5,7]	228 [5]											
Chrysler	Sebring	Console, aft of			226 [4]	226		387/516 [8]									
Chrysler	Town & Co...	Center stack				385 [6]		387/385 [8]			387/785						
Dodge	Avenger	Console, aft of				226		387/516 [8]	387/790/516		387/516						
Dodge	Caliber				226 [4]	226		387/516 [8]	387/790/516								
Dodge	Caravan	Center sta...				385 [6]		387/385 [8]			387/785						
Dodge	Challenger	...r stack				228			387/228 [10]	387/228			387/598				
Dodge	Charger			~~226~~ [3,5]	228 [5]				387/598								
Dodge	Dakota/Raider	Center stack			228 [5]				228	Mex Only							
Dodge	Dart	Center Stack									387/785						
Dodge	Durango	Console, aft	~~226~~ [4]	~~226~~ [2,4]	~~226~~ [2,4]	~~226~~ [2]	~~226~~ [2]		387/228 [10]	387/228		387/598					
Dodge	Journey	Console, fwd of shifter					385	387/385 [8]			387/385	387/385 +516PPT				387/821	
Dodge	Magnum	Center stack		~~226~~ [3,5]	228 [5]												
Dodge	Nitro	Console, aft of			227 [4]	227		387/516 [8]									
Dodge/Ram	...romaster City												387/802	+ 709 DLC Adapter			
Dodge/Ra	...(Big Van)											387/829 new 17_4		+ 709 DLC Adapter		387/785	
Dodge/Ram	Ram ...	CF Seat/console		226 [4]				387/453 [8]									
Dodge/Ram	Ram Pickup (2...	...t/console		226 [4]			226 [9]	387/453 [8]									
Dodge/Ram	Ram Pickup (1500)			226 [4]		226	453 [8]	387/453 [8]	387/453		387/785						
Dodge/Ram	Ram Van	Center Sta...									387/785						
Dodge/SRT	Viper	Center Stack									387/785						
Jeep	Commander	Console, aft		~~226~~ [3,5]	228 [5]												
Jeep	Compass/Patriot	Center stack			226 [4]	226		387/516 [8]	790/516		387/516		387/ *808 OR 819*				387/819
Jeep	Grand Cherokee	Console, aft of		~~226~~ [3,5]	228 [5]				387/228 [1]	387/228+	387/228	387/598					
Jeep	Liberty/Cherokee	Console, aft of				227		387/516 [8]			?	387/785					
Jeep	Renegade													387/819*			
Jeep	Wrangler	Console, aft of			227 [4]	227		387/546 [8]			387/546					387/821	

FCA EDR Evolution (under dev't)



Output New Look –

Removed multiple brake elements, added ABS/SC, moved strg to P1, hi res

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Pre-Crash Recorder Status	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full	Service Brake	Engine RPM	ABS Activity	Stability Control	Steering Input (deg)
-5.0	Complete	96 [155]	0.00	Off	3,878	No	On	1
-4.9	Complete	96 [154]	0.00	Off	3,835	No	On	-1
-4.8	Complete	96 [154]	0.00	Off	3,726	No	On	-5
-4.7	Complete	96 [154]	0.00	Off	3,621	No	On	-10
-4.6	Complete	96 [154]	0.00	Off	3,489	No	On	-18
-4.5	Complete	95 [154]	0.00	Off	3,344	No	On	-21
-4.4	Complete	95 [153]	0.00	Off	3,210	No	On	-20
-4.3	Complete	95 [153]	0.00	Off	3,109	No	On	-19
-4.2	Complete	95 [153]	0.00	Off	3,068	No	On	-19
-4.1	Complete	95 [152]	0.00	Off	3,108	No	On	-18
-4.0	Complete	95 [152]	0.00	Off	3,079	No	On	-18

Wheel RPM replaced by speed

See driven vs non driven, left vs right, ABS working…

Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	ABS MIL	Yaw Rate (deg/sec)	Wheel Speed LF (km/h)	Wheel Speed RF (km/h)	Wheel Speed LR (km/h)	Wheel Speed RR (km/h)
-5.0	Off	0.40	154.37	154.52	154.55	154.52
-4.9	Off	-0.48	154.28	154.16	154.25	154.25
-4.8	Off	-1.76	154.40	154.09	153.88	153.77
-4.7	Off	-2.80	154.28	153.95	153.88	153.67
-4.6	Off	-4.24	154.01	153.64	153.85	153.55
-4.5	Off	-6.80	153.98	153.16	153.80	153.19
-4.4	Off	-7.12	153.80	152.88	153.70	152.80
-4.3	Off	-5.84	153.49	152.67	153.37	152.62
-4.2	Off	-5.12	153.13	152.19	153.25	152.38
-4.1	Off	-4.64	152.80	152.04	152.91	151.91
-4.0	Off	-4.64	152.49	151.73	152.46	151.65
-3.9	Off	-4.00	152.13	151.55	152.13	151.31
-3.8	Off	-3.68	151.83	151.25	151.95	151.13

Current Gear & More @ 0.25 sec

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

Time Stamp (sec)	Pre-Crash Recorder Status	Raw Manifold Pressure (kPa)	PCM MIL	ETC Lamp	Current Gear	Reverse Gear Status
-5.00	Complete	20.00	Off	Off	D5	Not_Reverse
-4.75	Complete	20.00	Off	Off	D5	Not_Reverse
-4.50	Complete	21.60	Off	Off	D5	Not_Reverse
-4.25	Complete	23.20	Off	Off	D5	Not_Reverse
-4.00	Complete	23.20	Off	Off	D6	Not_Reverse
-3.75	Complete	23.20	Off	Off	D6	Not_Reverse
-3.50	Complete	23.20	Off	Off	D6	Not_Reverse
-3.25	Complete	23.20	Off	Off	D6	Not_Reverse
-3.00	Complete	22.40	Off	Off	D6	Not_Reverse
-2.75	Complete	21.60	Off	Off	D6	Not_Reverse
-2.50	Complete	21.60	Off	Off	D7	Not_Reverse
-2.25	Complete	21.60	Off	Off	D7	Not_Reverse
-2.00	Complete	21.60	Off	Off	D7	Not_Reverse
-1.75	Complete	21.60	Off	Off	D7	Not_Reverse

‹ 10.63 in ‹

Tire pressure @ 0.25 sec intervals

Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)

(the most recent sampled values are recorded prior to the event)

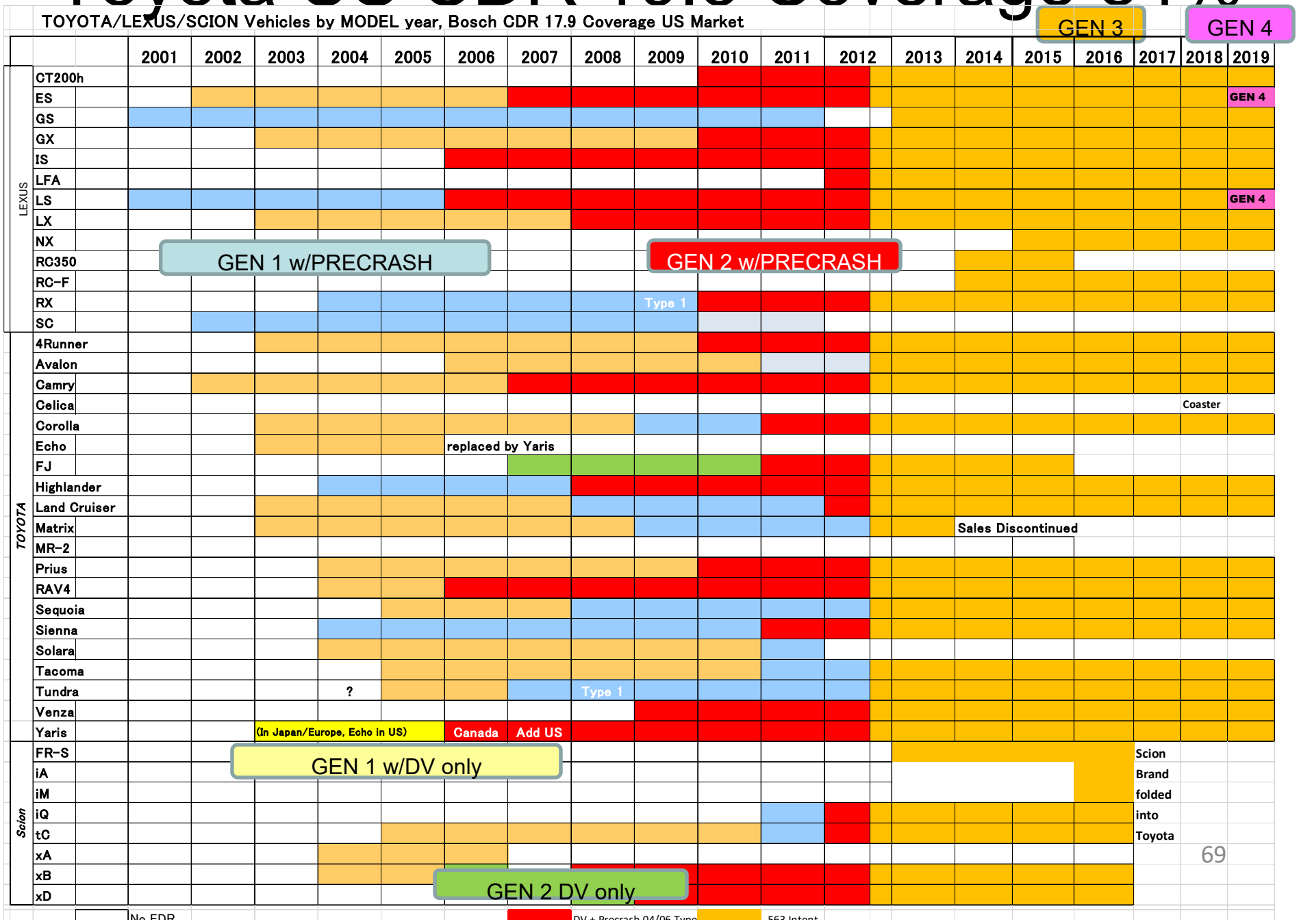
NEW

Time Stamp (sec)	Tire Pressure Position	Tire Pressure Status	Tire Pressure Value (PSI)	Cruise Control Lamp Status	Cruise Control Engaged Status	ACC On/Off	ACC Set Speed (MPH [km/h])
-5.00	LHF	Normal	34	Off	Not_Engaged	Off	0 [0.0]
-4.75	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.50	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.25	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-4.00	RHR	Normal	32	Off	Not_Engaged	Off	0 [0.0]
-3.75	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.50	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.25	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-3.00	LHR	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.75	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]
-2.50	RHF	Normal	33	Off	Not_Engaged	Off	0 [0.0]

Pedestrian Protection

1. Starting 2012 Fiat Freemont & Dodge Journey, ***In Europe only***, FCA has added a pedestrian protection feature to deploy a feature that pops the hood up slightly to make the angle better and less injurious to pedestrians.
2. It is a separate module with separate data and must be imaged separately thru the DLC or via cable 516.
3. Widespread in Europe, may come to US

Toyota US CDR 19.3 Coverage 84%



Toyota EDR Evolution

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
3 front 1 side 1 roll	2 front 2 side 2 roll		4 front 4 side 6 precrash
Record on 2g wakeup		Record on 5mph XDV, side 3.5g wake	Record on 5 X or Y DV
150ms XDV 5PC@1 + TRG	200ms XDV 5PC@1+TRG	250ms XDV 10PC@0.5 + TRG	250ms XY 10PC@0.5+TRG

Toyota EDR Evolution cont'd

GEN 1 00,02 EDR	GEN 2 04,06 EDR	GEN 3 12,13,15 EDR	GEN 4 17 EDR
70ms YDV@ACM+sats		12 YDV 70ms sats only 13 250ms ACM	250ms YDV
REFRESH 500 MS SPEED		REFRESH 24 MS SPEED	
78 MPH TOP, 2kph res	75 MPH TOP. 2kph res	124 MPH TOP 1 kph res	
Accel Pedal Volts		Accel Pedal % & Throttle %	

Toyota EDR Evolution cont'd

GEN 1
00,02 EDR

GEN 2
04,06 EDR

GEN 3
12,13,15 EDR

GEN 4
17 EDR

RPM to next lower 400

RPM to 100

No Key Cycles

Key Cycles

Time Between Events < 5 sec

Time Between Events <16 sec

Brk Oil Pres, Stab. Cont. Long Accel

Strg & Yaw

Gen 1 & 2 Pre Crash Data

5 Points a 1 sec intervals

6th point Sampled at TRG

Pre-Crash Data, -5 to 0 seconds (Most Recent, TRG 2)

Time (sec)	-4.2	-3.2	-2.2	-1.2	-0.2	0 (TRG)
Vehicle Speed (MPH [km/h])	12.4 [20]	8.7 [14]	5 [8]	2.5 [4]	1.2 [2]	1.2 [2]
Brake Switch	ON	ON	ON	ON	ON	ON
Accelerator Rate (V)	0.78	0.78	0.78	0.78	0.78	0.78
Engine RPM (RPM)	800	800	800	400	800	800

Caution: Sampling at TRG could have un-refreshed data in buffer

New Pre Crash data will only be recorded if 500ms have passed since the last event.

2018 Toyota Precrash Data

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4

Time (sec)	Vehicle Speed (MPH [km/h])	Accelerator Pedal, % Full (%)	Percentage of Engine Throttle (%)	Fuel Injection Quantity (mm^3/st)	Engine RPM (RPM)	Motor RPM (RPM)	Service Brake, ON/OFF
-5.00	29.8 [48]	49.5	24.5	Invalid	2,400	Invalid	OFF
-4.50	32.3 [52]	47.0	24.5	Invalid	2,500	Invalid	OFF
-4.00	34.2 [55]	38.0	20.0	Invalid	2,700	Invalid	OFF
-3.50	35.4 [57]	32.5	16.0	Invalid	2,800	Invalid	OFF
-3.00	36.7 [59]	26.5	13.5	Invalid	2,400	Invalid	OFF
-2.50	37.3 [60]	22.5	11.0	Invalid	2,400	Invalid	OFF
-2.00	37.3 [60]	19.0	8.5	Invalid	2,200	Invalid	OFF
-1.50	38.5 [62]	19.0	6.0	Invalid	1,900	Invalid	OFF
-1.00	37.9 [61]	18.0	5.5	Invalid	1,900	Invalid	OFF
-0.50	37.9 [61]	18.0	5.0	Invalid	1,800	Invalid	OFF
TRG(0)	36.0 [58]	100.0	26.0	Invalid	1,700	Invalid	OFF

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4

Time (sec)	ABS Control Status	BOS Control Status	Brake Oil Pressure (Mpa)	Longitudinal Acceleration , VSC Sensor (m/s^2)	Yaw Rate (deg/s)	Steering Input (degrees)	Shift Position
-5.00	OFF	OFF	0.00	1.579	-0.98	-4.5	D
-4.50	OFF	OFF	0.00	1.795	0.00	-1.5	D
-4.00	OFF	OFF	0.00	1.292	0.49	0.0	D
-3.50	OFF	OFF	0.00	0.861	0.00	-1.5	D
-3.00	OFF	OFF	0.00	0.718	0.00	-1.5	D
-2.50	OFF	OFF	0.00	-0.072	0.00	-1.5	D
-2.00	OFF	OFF	0.00	0.000	0.00	-1.5	D
-1.50	OFF	OFF	0.00	0.000	-0.49	-1.5	D
-1.00	OFF	OFF	0.00	0.287	0.00	-1.5	D
-0.50	OFF	OFF	0.00	-0.144	0.00	-1.5	D
TRG(0)	OFF	OFF	0.48	-1.507	61.00	-19.5	D

Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4

Time (sec)	Sequential Shift Range	Cruise Control Status	VSC Control Status	READY Signal	Drive Mode, Power Train	Drive Mode, Snow	Drive Mode, EV
-5.00	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid
-4.50	Undetermined	OFF	ON (enable)	Invalid	Normal	OFF	Invalid

2018 Pedestrian Protection

System Status at Pedestrian Event (1st Prior Event, TRG 1)

TRG Count (times)	1
TRG Count not for Pedestrian (times)	SNA
Ignition Cycle, Crash (times)	353
Odometer signal (miles [km])	1,195 [1,923]
Trip count (times)	392
Time count (msec)	1,283,900
Time count input system	Normal
Airbag Warning Lamp, On/Off	OFF
Pedestrian Warning Lamp, On/Off	OFF
Time from Previous TRG (msec)	32767 or greater
Time from Pre-Crash to TRG (msec)	300
Time between TRG not for Pedestrian and TRG for Pedestrian (msec)	SNA
TRG Establishment not for Pedestrian, within 5 sec.	ON
Freeze Signal	ON
Recording Status, Pre-Crash	Complete
Recording Status, Crash Info.	Complete
Recording Status, EDR Correlation Info.	Complete

Deployment Command Data (1st Prior Event, TRG 1)

Front Pop-Up Hood, Time to Deploy (msec)	3.0
Rear Pop-Up Hood, Time to Deploy (msec)	3.0
Airbag for Pedestrian Deployment, Time to Deploy (msec)	SNA

DTCs Present at Time of Event (1st Prior Event, TRG 1)

Recording Status, Diagnostic	Complete
Ignition Cycle Since DTC was Set (times)	0
Warning ON Time Since DTC was Set, for Pedestrian (min)	0
Diagnostic Trouble Codes	None

Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

Recording Status, Time Series Data	Complete
Time from TRG to Next Sample (msec)	1

Additional Freeze Frame Data 2013+

1. New "**Vehicle Control History**" feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, Camry, Corolla, Prius C, Prius V. 2016+ Avalon, Land Cruiser, Mirai, Prius, Tacoma, Scion iM, and Lexus ES, GS, LX, and RX. Likely in all now!
2. Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC. Now Lane Keep Assist, ICS, EDR too.
3. Accessible with Toyota Techstream service tool (or Nexiq interface or CDR900) and Toyota service subscription software (days, month, or annual) (TSS 2.0 now out, new undocumented trigger #25 EDR).
4. Data exports as CSV to Microsoft excel
5. Space in memory for MANY events – can show a history of erratic driver behavior
6. Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** "**Reconstructing Vehicle Dynamics from On-Board Event Data**" **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events)**

TOYOTA Vehicle Control History and TSS Crash Avoidance

		2013	2014	2015	2016	2017	2018	2019	2020
		CHART IS INCOMPLETE (especially Lexus) AS OF THIS PRINTING							
LEXUS	ES							GEN 4	
	GS	absense of TSS/LSS entries does not have signficance, they are not filled in yet							
	GX								
	IS								
	LC								
	LS							GEN 4	
	LX								
	NX								
	RC								
	RX								GEN 4
	UX								
TOYOTA	4Runner								
	86								
	Avalon				TSS		GEN4		
	CH-R								
	Camry						GEN4		
	Corolla					TSS		GEN4	
	GR Supra								
	Highlander					TSS			
	iM								
	Land Cruiser								
	Mirai								
	Prius				VCH/TSS				GEN4
	Prius C (city hybrid)				VCH/TSS				
	Prius Prime (plug in hybrid)					VCH/TSS			
	RAV4								
	Sequoia						VCH/TSS/GEN4		
	Sienna						VCH/TSS		
	Tacoma						TSS		
	Tundra						VCH/TSS/GEN4		
	Venza				out of production				
	Yaris					TSS	VCH		

Honda/Acura CDR 19.3 Covr'g 45%

Model	Location	Connector									
		2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
ACURA				Last 3 digits of Bosch CDR cable number							
RL	Under dashboard		547								
TL	Under dashboard		547								
TSX	Under dashboard		547								
ZDX	Under dashboard		547								
MDX	Under dashboard		547		387/789			387/810			
NSX	Under dashboard							387/789			
RDX	Under dashboard			547			387/789		387/825		
ILX	Under dashboard			547			387/789				
RLX	Under dashboard				387/789						
TLX	Under dashboard					387/789				387/810	
HONDA			2012	2013	2014	2015	2016	2017	2018	2019	2020
ACCORD	Under dashbord c			387/789			387/810		387/825		
CIVIC 2 DR	Under dashbord c		547				387/810				
CIVIC 4 DR	Under dashbord c		547	387/789			387/810				
CROSSTOUR	Under dashbord c		547								
CR-V	Under dashbord c		547				387/789	387/810			
CR-Z	Under dashbord c		547								
FCX Clarity	Under dashbord c							387/810			
FIT	Under dashbord c	EV	547			387/789		387/810			
HR-V							387/789		387/810		
INSIGHT	Under dashbord c		547							387/810	
ODYSSEY	Under dashbord c		547		387/789			387/810			
PASSPORT	Under dashbord c									387/810	
PILOT	Under dashbord c		547				387/789	387/810			
RIDGELINE	Under dashbord c		547					387/810			

2011- NO time series EDR But may have Delta V single value

Honda only assists law enforce -ment voluntar ily

The source data for this chart is the Bosch CDR help file for Version 19.3

Honda Precrash

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Same unless brake or stability control override

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	PCM Derived Accelerator Pedal Position, % full	Service Brake (On, Off)	ABS Activity (On, Off)	Stability Control (On, Off, Engaged)	Steering Input (deg)	Engine RPM	Accelerator Pedal Position, % full
-5.0	105 [169]	93	Off	Off	On Non-Engaged	RTFDL Pos=Left	5,800	93
-4.5	106 [170]	93	Off	Off	On Non-Engaged		5,800	93
-4.0	106 [170]	93	Off	Off	On Non-Engaged		5,900	93
-3.5	107 [172]	93	Off	Off	On Non-Engaged	0	5,900	93
-3.0	108 [174]	26	Off	Off	On Non-Engaged	0	5,300	26
-2.5	108 [174]	85	Off	Off	On Non-Engaged	0	4,600	85
-2.0	108 [174]	0	On	Off	On Non-Engaged	-20	4,600	0
-1.5	108 [174]	0	On	Off	On Non-Engaged	-5	4,400	0
-1.0	103 [166]	0	On	On	On Non-Engaged	-20	3,900	0
-0.5	103 [166]	0	On	On	On Non-Engaged	0	3,100	0
0.0	90 [145]	0	On	On	On Non-Engaged	10 ?	2,400	0

NISSAN EDR's 19.3 (CDR 51%)

Model	Location	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR cable number																
INFINITI																
G CONVERTIBLE	TUNNEL BETWEEN SEATS				*		Jul-12		780							
G COUPE	TUNNEL BETWEEN SEATS		*		Yellow = Proprietary Consult Tool Only				780							
G SEDAN	TUNNEL BETWEEN SEATS		*		No direct to module cables		Jul-12		780							
M	TUNNEL BETWEEN SEATS								780							
Q30													780	780	780	780
Q40	TUNNEL BETWEEN SEATS										780					
Q50	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
Q60	TUNNEL BETWEEN SEATS								780	780	780		780	780	780	780
Q70	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	
EX	TUNNEL BETWEEN SEATS				Dealers do NOT have special software				780							
FX	TUNNEL BETWEEN SEATS				required to read EDR				780							
JX35	TUNNEL BETWEEN SEATS								780							
QX50-80	TUNNEL BETWEEN SEATS								780	780	780	780	780	50,60,80	50,60,80	50,60
QX30	TUNNEL BETWEEN SEATS												598	598	598	598
NISSAN		2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
370Z	TUNNEL BETWEEN SEATS		*				Aug-12		780	780	780	780	780	780	780	780
ALTIMA COUPE	TUNNEL BETWEEN SEATS			*					780	INCLUDED IN SEDAN BELOW						
ALTIMA SEDAN	TUNNEL BETWEEN SEATS		*		Yellow = Proprietary Consult Tool Only				780	780	780	780	780	780	780	780
ARMADA	TUNNEL BETWEEN SEATS	*			No direct to module cables				780	780	780	?	780	780	780	780
CUBE	TUNNEL BETWEEN SEATS				*				780	780						
FRONTIER	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780	780	780	780	780	
GT-R	TUNNEL BETWEEN SEATS				*					780	780	780	780	780	780	780
JUKE	TUNNEL BETWEEN SEATS								780	780	780	780	780			
KICKS														780	780	780
LEAF	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
MAXIMA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780	780	780
MICRA	Check Manual										780	780	780	780	780	
MURANO	TUNNEL BETWEEN SEATS		*		*				780	780	780	780	780	780	780	780
MURANO CROSS C	TUNNEL BETWEEN SEATS								included in above							
NV200	TUNNEL BETWEEN SEATS								780	780	780	780	780	780	780	780
PATHFINDER	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780	780	780
QUEST	CTR CONSOLE NEAR FRONT		*						780	780	780	780	780			
ROGUE, R SPORT	Check Manual			*					780	780	780	780	780	+Qashqai	780	780
SENTRA	TUNNEL BETWEEN SEATS		*						780	780	780	780	780	780	780	780
TITAN	TUNNEL BETWEEN SEATS	*							780	780	780	780	780	780	780	780
VERSA SEDAN	TUNNEL BETWEEN SEATS		*						780	780	780	780		780	780	780
VERSA NOTE HB	TUNNEL BETWEEN SEATS						Sep-12		780	780	780	780	780	780	780	
XTERRA	TUNNEL BETWEEN SEATS	*					Oct-12		780	780	780					

Yellow area not covered by CDR

Nissan Pre-Crash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most recent sampled values are recorded prior to the event)

Time Stamp (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % full	Engine RPM	Motor RPM	Service Brake (On, Off)	Steering Input (deg)
-5.0	24 [39]	16	2650	0	Off (Brake Not Activated)	-32
-4.5	23 [37]	11.5	1750	0	Off (Brake Not Activated)	-2
-4.0	21 [34]	9	1200	0	On (Brake Activated)	28
-3.5	15 [24]	9	1150	0	On (Brake Activated)	14
-3.0	13 [21]	4	950	0	On (Brake Activated)	4
-2.5	11 [17]	4	950	0	On (Brake Activated)	4
-2.0	5 [8]	4	900	0	On (Brake Activated)	-48
-1.5	3 [5]	4	850	0	On (Brake Activated)	-98
-1.0	1 [2]	4	850	0	On (Brake Activated)	-160
-0.5	3 [5]	10.5	1100	0	Off (Brake Not Activated)	-194
0.0	6 [9]	16	1150	0	Off (Brake Not Activated)	-218

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

Mazda CDR 19.3 Coverage 48%

Model	Location	Connector	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR cable number shown														
MAZDA		F-00K-108-			All Mazda use 387 adapter for direct to module									
CX-3	TUNNEL BELOW IP	Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers. Mostly Delta V.								812				
Tribute/CX-5	TUNNEL AFT SHIFTER						779			811	812			
CX-9	TUNNEL BELOW IP				778					811				
Mazda 2	TUNNEL AFT SHIFTER				792				not listed	812	not listed			
Mazda 3	TUNNEL AFT SHIFTER					779				811			843	
Mazda 5	TUNNEL BELOW IP					778					not listed			
Mazda 6	TUNNEL BELOW IP				792			779		811		831		
MX-5 Miatta	TUNNEL BELOW IP						no coverage			812				
CX-30														
Memo: The Mazda Tribute was sister to the Ford		F-00K-108-												

The source data for this chart is the Bosch CDR help file for Version 19.3
Always consult the latest help file for the most accurate coverage

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Typical Mazda Precrash Data

Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)

(the most rec t sampled values are recorded prior to the event)

Time Stamp (sec)	Previous 22	Speed, Vehicle Indicated (MPH [km/h])	Engine Throttle, % full	Service Brake (On, Off)
-5.0	19	19 [31]	0	On
-4.5	17	17 [27]	0	On
-4.0	14	14 [23]	0	On
-3.5	12	12 [19]	0	On
-3.0	11	11 [17]	25	Off
-2.5	10	10 [16]	61	Off
-2.0	11	11 [17]	63	Off
-1.5	13	13 [21]	64	Off
-1.0	16	16 [26]	64	Off
-0.5	18	17 [27]	0	Off
0.0		4 [6]	0	Off

Minimum Regulatory Content

Recording threshold <5mph

This is a second event – last data pt at impact

Volvo CDR 19.3 Coverage 32%

Model	Location	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR cable number sh												
VOLVO		All Volvo use 387 Adapter for DTM images										
S40												
S60	CTR TUNNEL UNDER CONSOLE		799							500/830		
S80, S80L	CTR TUNNEL UNDER CONSOLE											
S90	CTR TUNNEL UNDER CONSOLE								816			
V40	CTR TUNNEL UNDER CONSOLE								816 Euro			
V60	CTR TUNNEL UNDER CONSOLE		799							500/830		
V70	CTR TUNNEL UNDER CONSOLE				799							
V90	CTR TUNNEL UNDER CONSOLE								816			
XC40	CTR TUNNEL UNDER CONSOLE									500/830		
XC60	CTR TUNNEL UNDER CONSOLE				799							
XC70	CTR TUNNEL UNDER CONSOLE				799							
XC90	CTR TUNNEL UNDER CONSOLE							816				

Old Volvo DV only Law Enf only call Volvo

The source data for this chart is the Bosch CDR help file for Version 19.3
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2016+ XC90 Sample Volvo File

(vs. 2013 Volvo with minimum data)

Pre-Crash Data -1 Sec (Event Record 1)

Ignition Cycle, Crash	2,764
Safety Belt Status, Driver	On, Belted
Safety Belt Status, Passenger	On, Belted
Frontal Airbag Warning Lamp	Off
Frontal Airbag Suppression Switch Status, Front Passenger	On
Seat Track Position Switch, Foremost, Status, Driver	Not Equipped
Seat Track Position Switch, Foremost, Status, Front Passenger	Not Equipped
Occupant Size Right Front Passenger Child	Not Equipped

Added Data for XC90

Pre-Crash -5 to 0 sec (Event Record 1)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Service Brake (On, Off)	Steering input (%)	ABS Activity	Stability control status
-5.0	43.5 [70.0]	100.0	Off	0.0	Off	On
-4.5	47.8 [77.0]	100.0	Off	0.0	Off	On
-4.0	52.2 [84.0]	100.0	Off	2.0	Off	On
-3.5	55.9 [90.0]	100.0	Off	3.0	Off	On
-3.0	58.4 [94.0]	100.0	Off	3.0	Off	On
-2.5	61.5 [99.0]	100.0	Off	11.0	Off	Engaged
-2.0	62.8 [101.0]	100.0	Off	6.0	Off	Engaged
-1.5	65.2 [105.0]	100.0	Off	-9.0	Off	Engaged
-1.0	67.1 [108.0]	100.0	Off	-8.0	Off	Engaged
-0.5	69.6 [112.0]	100.0	Off	-9.0	Off	Engaged
0.0	70.2 [113.0]	100.0	Off	-13.0	Off	Engaged

BMW CDR US 19.3 Coverage 42%

See help file for Rolls Royce

Model	Location	2012	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR adapter/cable number shown										
BMW	**All BMW require 387 Adapter for DTM**									
1 Series										
2 Series	Driver side, behind IP, rt of strg			796						
3 Series	Driver side, behind IP, rt of strg		796							
Active Hybrid 3	Passenger side, behind the glove box		796							
4 Series	Driver side, behind IP, rt of strg			796						
5 Series	Passenger side, behind the glove box		796					500/822		
6 Series	Passenger side, behind the glove box		796							
6 Series GT	See Service Manual								500/822	
7 Series	Passenger side, behind the glove box		796			500/796	500/822			
8 Series	Passenger side, behind the glove box								500/822	
X3 xDrive	Passenger side, behind the glove box		796							
X5	Center Tunnel, Between Seats			796					500/822	
X1, X2	Center Tunnel, Between Seats				798	500/798	500/807		also X2	
X3, X4	See Help File - Various				796			500/822		
X6	Center Tunnel, Between Seats				796					
X7	See Service Manual								500/822	
Z4	Center Tunnel, Under Dash				798				500/822	
i3, i8	See Service Manual			500/807						

Europe only not covered

No “back model” coverage

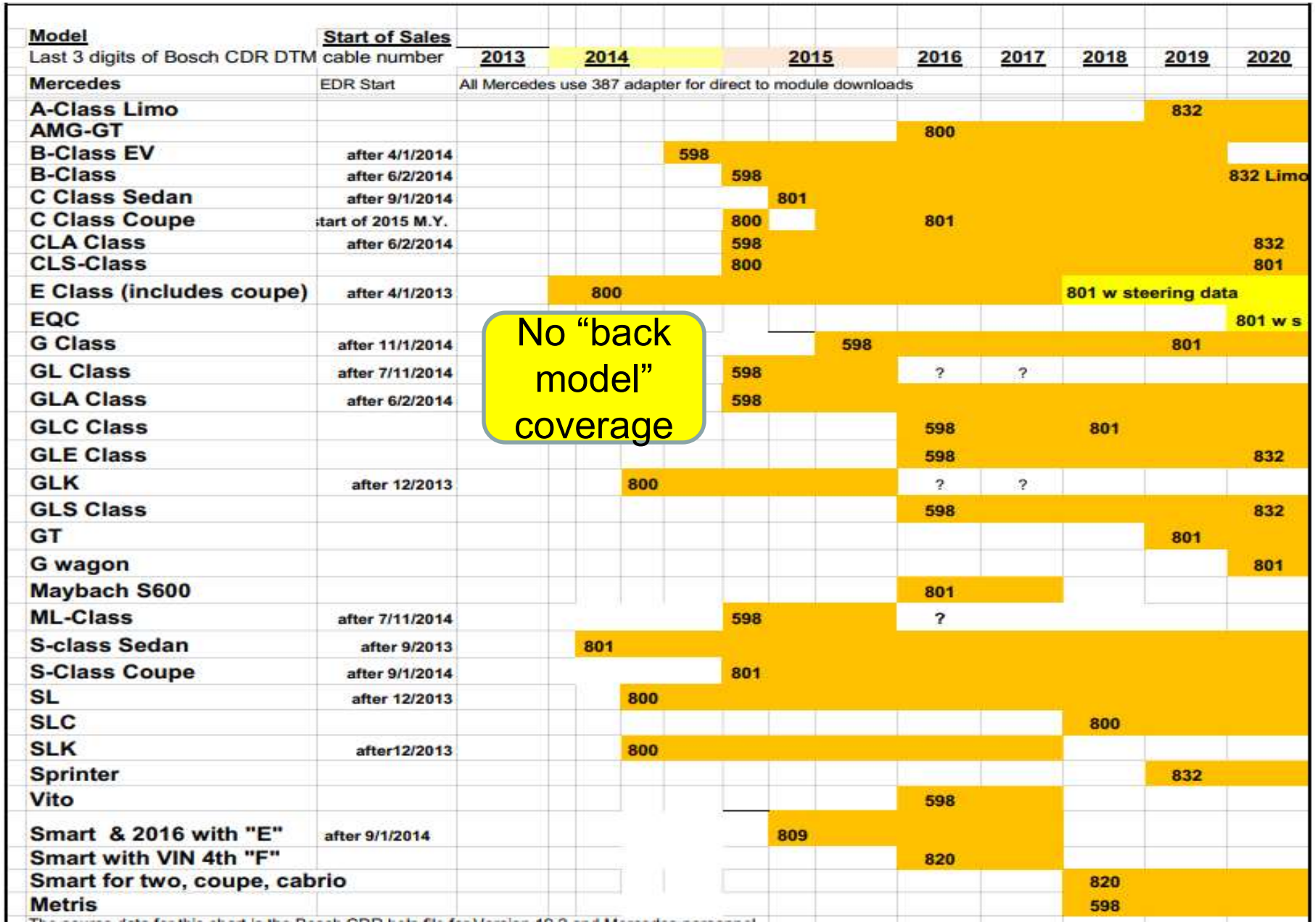
500=flex ray

BMW Sample Pre-Crash

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Engine RPM	Steering Input (deg)	Service Brake, On/Off	ABS Activity (Engaged, Non-engaged)	Stability Control (On Engaged, Non-engaged)
-5.0	55 [89]	16	1500	0	Off	No ABS Activity	Non-engaged
-4.5	55 [89]	17	1500	0	Off	No ABS Activity	Non-engaged
-4.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.5	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-3.0	55 [89]	19	1500	0	Off	No ABS Activity	Non-engaged
-2.5	55 [89]	8	1500	0	Off	No ABS Activity	Non-engaged
-2.0	55 [89]	7	1500	0	Off	No ABS Activity	Non-engaged
-1.5	55 [89]	14	1500	0	Off	No ABS Activity	Non-engaged
-1.0	55 [88]	19	1500	0	Off	No ABS Activity	Non-engaged
-0.5	55 [89]	0	1500	0	Off	No ABS Activity	Non-engaged
0.0	45 [73]	0	1300	10	On	ABS Activity	Non-engaged

Mercedes CDR Coverage 36%



Model	Start of Sales	2013	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR DTM cable number									
Mercedes	EDR Start	All Mercedes use 387 adapter for direct to module downloads							
A-Class Limo								832	
AMG-GT					800				
B-Class EV	after 4/1/2014		598						
B-Class	after 6/2/2014			598					832 Limo
C Class Sedan	after 9/1/2014			801					
C Class Coupe	tart of 2015 M.Y.			800	801				
CLA Class	after 6/2/2014			598					832
CLS-Class				800					801
E Class (includes coupe)	after 4/1/2013		800				801 w steering data		
EQC									801 w s
G Class	after 11/1/2014			598				801	
GL Class	after 7/11/2014			598	?	?			
GLA Class	after 6/2/2014			598					
GLC Class					598		801		
GLE Class					598				832
GLK	after 12/2013		800		?	?			
GLS Class					598				832
GT								801	
G wagon									801
Maybach S600					801				
ML-Class	after 7/11/2014			598	?				
S-class Sedan	after 9/2013		801						
S-Class Coupe	after 9/1/2014			801					
SL	after 12/2013		800						
SLC							800		
SLK	after12/2013		800						
Sprinter								832	
Vito					598				
Smart & 2016 with "E"	after 9/1/2014			809					
Smart with VIN 4th "F"					820				
Smart for two, coupe, cabrio							820		
Metris							598		

No “back model” coverage

Mercedes Pre-Crash

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal (%)	Service Brake Activation
-5.0	48 [77]	0	On
-4.5	47 [76]	16	Off
-4.0	48 [77]	15	Off
-3.5	48 [77]	20	Off
-3.0	48 [77]	20	Off
-2.5	48 [78]	21	Off
-2.0	49 [79]	0	On
-1.5	46 [74]	0	On
-1.0	39 [63]	0	On
-0.5	31 [50]	0	On
0.0	25 [40]	0	On

VW/Audi 19.2 US Coverage 34%

See help file for Rolls Royce

Model	Location	2009-13	2014	2015	2016	2017	2018	2019	2020
Last 3 digits of Bosch CDR adapter/cable numbe									
All VW/Audi Require the 387 Adapter to read DTM			Gold = US only (Canada not covered)				Yellow=Worldwide		
VW									
Arteon	UNDER CENTER STACK						805		
Atlas	UNDER CENTER STACK						805		
Beetle	UNDER CENTER STACK			804			Canada added		
CC	UNDER CENTER STACK			804					
Crafter	UNDER CENTER STACK						805		
EOS	UNDER CENTER STACK		806						
Golf, eGolf	UNDER CENTER STACK			805			Worldwide		
Jetta	UNDER CENTER STACK			804			805		
Passat	UNDER CENTER STACK			804				500/826	
Polo	UNDER CENTER STACK						805		
Routan	UNDER CENTER STACK	785							
Sportwagen + Alltrack				Golf platform, not called out separtely in help file					
T-Roc	UNDER CENTER STACK							805	
T-Cross	UNDER CENTER STACK						805		
Tiguan	UNDER CENTER STACK			804		805			
Touareg	UNDER GEAR SELECTOR			804					
Touran	UNDER CENTER STACK						805		
AUDI				2015	2016	2017	2018	2019	2020
A1	Under Center Stack							805	
A3,S3	Under Center Stack			805					842
A4, allroad	Tunnel between 1st/2nd row seats			804		500/813			
A5, S5, RS5	Tunnel between 1st/2nd row seats			804			500/813		
A6, S6	Tunnel between 1st/2nd row seats			804			500/826		
A7, S7, RS7	Tunnel between 1st/2nd row seats			804			500/826		
A8, S8	Tunnel between 1st/2nd row seats			804			500/826		
Q2	Under Center Stack						805		
Q3	Under Center Stack			804				805	
Q5, SQ5	Tunnel between 1st/2nd row seats			804			500/813		
Q7	Under Center Stack					500/813			500/826
Q8	Under Center Stack						500/826		
R8	Under Center Stack					804			
TT, TTS				not covered at this time					
eTron	Under Center Stack							500/826	

Chrysler Minivan Sister

No "back model" coverage

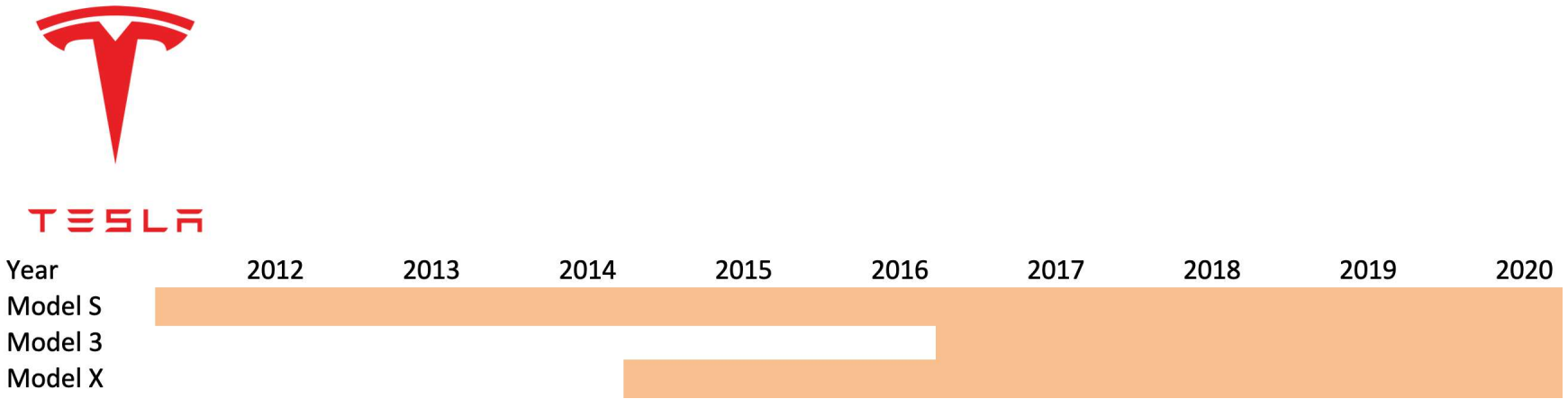
VW/Audi Pre-Crash Data

Pre-Crash Data -5 to 0 sec (Record 2)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal (%)	Service Brake Activation
-5.0	35 [56]	13	Off
-4.5	35 [56]	14	Off
-4.0	35 [57]	0	Off
-3.5	35 [57]	0	Off
-3.0	35 [57]	0	Off
-2.5	35 [56]	0	Off
-2.0	34 [55]	0	Off
-1.5	34 [54]	0	Off
-1.0	32 [52]	0	Off
-0.5	32 [51]	0	Off
0.0	29 [47]	2	Off

Tesla EDR

Requires Tesla Specific Tool $1200
More information is available on their website https://edr.tesla.com



What about OTHER data Tesla Data Records?

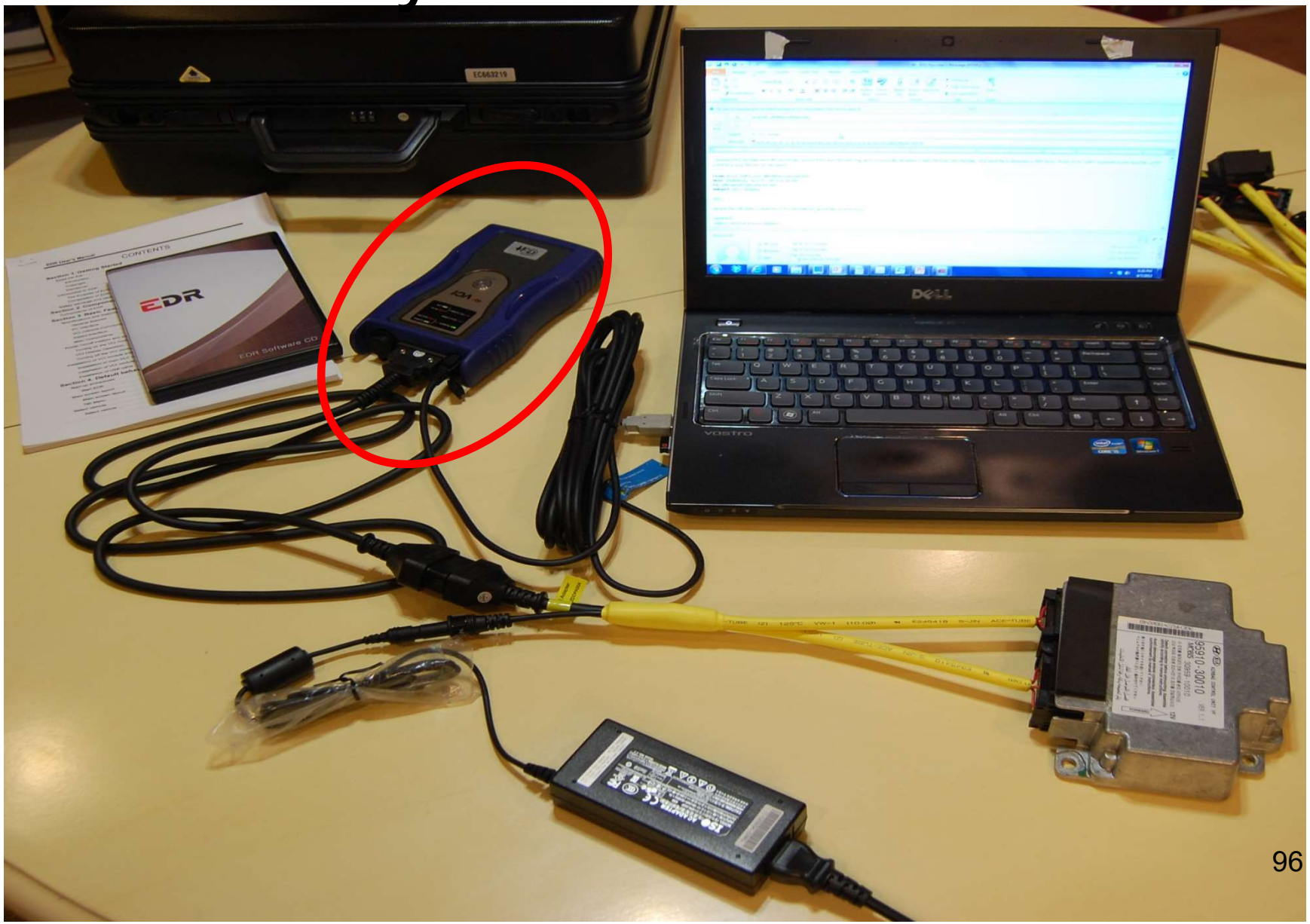
1. Tesla records everything the car does onto an SD card and beams it up to the cloud where Tesla stores it
2. Modern Tesla have recorded video that can be uploaded to Tesla
3. This data can be retrieved from Tesla
4. Contact legal@tesla.com



Hyundai/Kia Tools

- There is no DLC only kit. Hyundai kit comes with 13 DTM cables thru #24, Kia with 15 DTM cables thru #23. 10 Cables are in both. New cables cost $250. #25-30 for 2019-2021's not yet included.
- Hyundai: Call GIT America Tool Company order Part G0Z**H**DMN001 ***$6450 does not include software.***
- GIT now also sells Kia tool, (was Snap On),order part GIT0Z**K**DMN001 ***$6950 + tax+ship***
- Software updates **$495/yr** per brand from GIT Tool Company. Software V#35 Hyundia/ V#34 Kia released 3/10/2020. New $1120 CAN-FD adapter for GV80.
- Dan Portillo dportillo@gitamerica.com(888) 549-4977
- Different version of tool for Korea and non-US markets.

The Hyundai EDR Reader



Kia/Hyundai 2013+ Models V34/35

Model		Cable # for Model Year shown below							
KIA	**Vehicle Size, Description**	**2013***	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**
CADENZA (Azera sister)	112.0" WB		19(VG)	19	19	13(YG)	13	13	
FORTE	104.3"WB, compact	10 (TD)	6 (YD)	6	6	12(YDm)	12	21(BDm)	21
K900 (Quoris)	119.9" luxury sedan			11 (KH)	11	11	11	21(RJ)	21
NIRO	106.3", subcom crossover					21(DE HEV)	21	21	21
OPTIMA	110.0" WB sedan	4 (TF/QF)	4	4	14 (JF,JFA)*	14	14	14	14
OPTIMA HYBRID	110.0" WB sedan	(TF HEV)	4	4	4,23(JFHEV)	23(JF PHEV)	23	23	23
RIO (UB)	94.5"WB, compact	8(UB)	8	8	8	8	21 (SC)	21	21
RONDO (Canada only 2011 MY+)	106.3", MPV	(not officially supported since non-US, read as a Kia Rio)							
SEDONA	113.8/118.9" WB minivan	No	10(VQ)	12(YP)	12	12	12	12	12
SELTOS	103.5" crossover								
SORENTO	106.3" WB, crossover	3(XM)	3,7(XM)	3,7	13 (UMA)	13	13	13	13
SOUL	100.4" WB	1(AM)	12(PS)	12(PSEV)	12	12	12	12, 26EV	26(SK3)
SPORTAGE	103.9" WB, crossover	2(SL)	2	2	2,13(QL)	13	13	13	13
STINGER	114.4 (sister to G70)						21(CK)	21	21
TELLURIDE	114.2 crossover								25(ON)
HYUNDAI		**2013***	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**
ACCENT	94.5"WB, compact	5(RB)	5	5	5	5	21(HC)	21	21
AZERA	109.4"WB, sedan	7(HG)	7	7	7	7			
ELANTRA Coupe, Sedan	104.3"WB, compact	8(JK/UD/MD)	8	8	8	21(AD, ADA)	21	21	21
ELANTRA GT (aka "i30")	former Elantra Touring	6,12(GD)	6,12	6,12	6,12	6,12	21(DT)	21	21
EQUUS (VI)(stretch genesis)	119.9" WB, luxury sedan	9(VI)	9,11(VI)	9,11	9,11				
GENESIS (Coupe , Sedan)	115.6" WB, luxury sedan	9(BK, BH)	9	9BK,13(DH)	9,13				
GENESIS G80	118.5 luxury sedan					13(DH)	13	13	13
GENESIS G90	124.4 luxury sedan					13(HI)	13	13	13
GENESIS G70	111.6 compact exec sedan							21(IK)	21
IONIQ EV and HEV	106.3 compact EV					21(AE)	21	21	21
KONA (i20, Rio sister)	102.4 crossover						24 (OS)	24	24
KONA EV								27 (OS EV)	27
NEXO	109.8 crossover fuel cell							26 (FE)	26
PALISADE (santa fe)	114.2 crossover								25(LX2)
SANTE FE	106.3-112.2, crossover	7(NC,DMA)	7	7	7	7	7	7,25 (TMA)	25
SONATA	110.0" WB sedan	4(YF YF HEV)	4	4,14 (LFA)	14 (LF HEV)	14	14	14	25(DN8A)
TUCSON	103.9" WB, crossover	2(LM)	2	2(LM FCEV)	2,13(TL)	2,13	13	13	13
VELOSTER	104.3" sporty coupe	5(FS)	5	5	5	5	5	14 (JS)	14
VELOSTER N	104.3 nex gen							28(JSN)	28
VENUE	99.2" crossover								30(QX)

Typical Kia/Hyundai Precrash Data
Accelerator pedal added in 2014

Pre-Crash Information (-5 ~ 0 sec)

All metric!

Time (sec)	Vehicle Speed [kph]	Engine RPM [rpm]	Engine Throttle [%]	Service Brake [on/off]	ABS Activity [on/off]	Stability Control [on/off/engaged]	Steering Input [degree]
-5.0	100	1900	10	OFF	OFF	on	0
-4.5	100	1900	10	OFF	OFF	on	0
-4.0	101	1900	9	OFF	OFF	on	0
-3.5	101	1900	9	OFF	OFF	on	0
-3.0	101	1900	9	OFF	OFF	on	0
-2.5	101	1900	9	OFF	OFF	on	0
-2.0	101	1900	9	OFF	OFF	on	0
-1.5	101	1900	9	OFF	OFF	on	0
-1.0	101	1900	9	OFF	OFF	on	0
-0.5	101	1900	9	OFF	OFF	on	0
0.0	101	1900	9	OFF	OFF	on	0

Subaru EDR Availability

CDR900

Hitachi HDS-3000

Denso DSTi Interface

SSM3 Software

SSM4

CDR 19.1 on some

Model / Year	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
SUBARU		<----Yellow=Old Hitachi SSM3 or new Denso w SSM4				Blue = SSM4 ONLY New Denso Tool				
Ascent									CDR 835	
BRZ (Scion FRS sister)		Japan	Yes SSM3	Yes	Yes	Yes	SSM4	Yes	Yes	
			For BRZ you can use CDR - select Toyota as Brand but enter Subaru VIN - it will work							
Impreza	No	Yes SSM3	Yes	Yes	Yes	SSM4	Yes	Yes	Yes	
WRX/STI	No	No	No	No	Yes SSM3	SSM4	Yes	Yes	Yes	
Legacy	No	No	Yes SSM3	Yes	CDR 836 Bosch ECU					
Outback	No	No	Yes SSM3	Yes	CDR 836 Bosch ECU					
Forester	No	No	Japan	Yes SSM3	Yes	Yes	SSM4	Yes	CDR 835	
Tribeca	No	No	No	No	Discontinued					
Crosstrek	n/a	n/a	Yes SSM3	Yes	Yes	SSM4	Yes	Yes	Yes	

CDR 19.4 is expected to cover remaining 2012+

Typical Bosch CDR Subaru Output of Pre-Crash Data

Pre-Crash Data -5 to 0 Sec (Record 1)

Time (sec)	-5.0	-4.5	-4.0	-3.5	-3.0	-2.5	-2.0	-1.5	-1.0	-0.5	0.0
Speed, Vehicle Indicated (MPH [km/h])	59.0 [95]	59.0 [95]	59.7 [96]	59.7 [96]	59.7 [96]	59.7 [96]	60.3 [97]	60.3 [97]	55.9 [90]	44.1 [71]	43.5 [70]
Accelerator Pedal, % Full (%)	14	14	14	11	10	14	11	0	0	0	0
Service Brake, On/Off	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
Engine RPM (RPM)	1,600	1,600	1,600	1,600	1,600	1,600	1,600	1,600	1,400	1,200	1,200
ABS Activity	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
Stability Control (On, Off, Engaged)	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON	ON
Steering Input (deg)	-7.5	-7.5	-7.5	-7.5	-5.0	-2.5	-2.5	-22.5	-7.5	-15.0	-17.5

Mitsubishi EDR's
Some use CDR, some use private tool

Model	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2
Eclipse (PS Platform)	Part MU141170 4/2005 to 8/2011		Proprietary EDR Tool		Production ends 8/2011								
Eclipse Spyder Conv.		?	Data in deployments only?										
Eclipse Cross (SUV)										837 Bosch CDR			
Endeavor SUV (PS Platform)		?			Production ends 8/2011 replaced by Outlander								
Galant (PS Platform)		?				Production ends 8/2012							
iMiEV		Not sold in US				No gateway module					?		
Lancer		2.3 sec@0.1 deploys only		Part 288 11-12 read w CDR as Caliber		ETACS gateway. Part A307-308 thru 2015					Production Ends		
Lancer Evolution		?	or send to Mitsu										
Lancer Sportback		?											
Mirage (subcompact)		Not in production				No gateway module							
Outlander SUV (GF platform) & PHEV		A208 Read w/CDR as Caliber		A208/A304 mix. A208 5 sec@0.1. Deploy only. Read as 09 Caliber.		ETACS gateway	838 Bosch CDR						
Outlander Sport, RVR, AS				838 Bosch CDR									
Raider pickup (Dodge Dakota sister)		Bosch CDR supported cable 228 (2007 -2010)		Not in production				Bosch cables 839 and 840 released in anticipation of future coverage - Mirage & Lancer??					

Typical Mitsubishi CDR Data

Pre-Crash Data -5 to 0 Sec (Record 1)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Service Brake, On/Off	Engine RPM (RPM)	Steering Input (deg)
-5.0	3.1 [5]	0	On	797	4.5
-4.5	2.5 [4]	0	On	751	4.5
-4.0	1.9 [3]	0	On	648	4.0
-3.5	1.2 [2]	0	On	593	4.0
-3.0	0.6 [1]	0	On	641	4.0
-2.5	0.6 [1]	60	Off	719	4.5
-2.0	1.2 [2]	71	Off	1,297	14.0
-1.5	4.3 [7]	75	Off	1,840	46.5
-1.0	7.5 [12]	77	Off	2,010	90.0
-0.5	9.9 [16]	76	Off	2,180	124.5
0.0	11.2 [18]	72	Off	2,206	146.0

- +Long & Lat Delta V and Acceleration
- +Roll Angle -1 to +5 sec @ 0.1 sec interval

Jaguar/Land Rover

1. Jag/LR have EDR's, they are read with a dedicated laptop and software. Tool only retrieves hex data, must be sent to England for interp. Only Robert Willer in FL has the tool (and he will help others).
2. Requests to insur@jaguarlandrover.com
3. **They are now referring requesters to <u>suppliers.</u>** (call to find out who supplier is)
4. Some Mid 2000's have ***PCM data*** (when Jag/LR was owned by Ford) but *not spoofable* with CDR.

Summary – Module 1

1. US leads world in EDR application
2. Early EDRs for mfr benefit
3. US Regulation standardizes min data, requires public tool, changes purpose of EDR to effective crash recon
4. Bosch CDR tool most widely used, but several additional tools to get all mfrs
5. Data varies widely by manufacturer, model and model year